

Thermodynamic Quantum Logic Gates: From the Autonomous Neuron to Floquet–Buffered Non–Markovian Operation

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Abstract

This document consolidates the theory, derivations, numerical reconstructions, non–Markovian tensor–network formulation, and Floquet buffer architecture developed for the thermodynamic–neuron project into a single reference. Part I (Sparsho) is a self–contained, build–up derivation: it starts from a single qubit in a heat bath, constructs the open–system master equation and shows explicitly how the system–bath interaction is absorbed into the dissipator, introduces the reset model, builds the virtual qubit and the resonant three–body interaction, assembles the autonomous NOT gate and the general linearly–separable neuron, reproduces the baseline logic numerically, and then extends the treatment beyond weak coupling through the Nakajima–Zwanzig equation, the influence functional, and process–tensor (PT–MPO/TEMPO) methods, culminating in the Floquet–buffer architecture and its robustness analysis. All approximations are collected in a single table (Table 1). Part II (Abhishek) contains the rigorous periodically–driven open–system formulation, kept intact.

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Part I

Sparsho: From the Single Qubit to Floquet–Buffered Thermodynamic Logic

1 Introduction: thermodynamics of computation

Thermodynamic computing studies the energetic and entropic cost of information processing, and asks whether physical relaxation can be used *directly* as a computational resource. In ordinary digital logic, robustness is obtained by engineering large barriers between stable states and by externally clocking transistor networks. In autonomous quantum thermal machines the situation is reversed: computation is performed by relaxation itself. Logical inputs are encoded into thermodynamic parameters (the inverse temperatures of heat baths), heat currents process those inputs, and the final state of an output degree of freedom is decoded as a logical value. This viewpoint continues the thermodynamics-of-information tradition of Landauer and Bennett and the quantum thermal-machine programme in which autonomous devices and *virtual temperatures* are operational resources.

The appeal of the autonomous model is that it needs no pre-programmed sequence of unitary gates. Its fragility is that the usual thermodynamic accounting depends on three conditions: weak system–bath coupling, rapid decay of bath correlations, and a clean separation between system timescales and reservoir memory. This part builds the model from the ground up so that every one of those conditions is visible and labelled (they are collected in Table 1), reproduces the baseline logic numerically, and then asks what must change when the conditions fail — which is precisely where the Floquet buffer and the non-Markovian machinery enter.

Scope of Part I. Sections 2–3.6 build the open-system tools; Sections 4–6 build the neuron; Section 7 reproduces the baseline logic numerically; Sections 8–9 extend the treatment beyond weak coupling and introduce the Floquet buffer; and Section 13 presents the robustness programme. Part II then gives the rigorous driven-dissipative formulation.

Derivation map for the supervisor

The document is intentionally written as a derivation trail rather than as a short paper. The following map states what each major block proves, where the physical idea enters, and why the step is needed.

Block	What is derived	Why it is needed
Single qubit	The Gibbs state $\tau(\beta)$ and the population $g(\beta\epsilon) = 1/(1 + e^{\beta\epsilon})$.	This is the thermal sigmoid from which the neuron activation function ultimately comes.
Open system derivation	The system–bath Hamiltonian is projected to a reduced master equation; the interaction H_{SB} becomes rates, jumps, Lamb shifts, and memory kernels.	This answers where the bath Hamiltonian and interaction have gone in the effective equations.
Reset model	The reset dissipator, heat current, entropy production, and the precise approximations that make them thermodynamically consistent.	This is the Markovian computational baseline used in the original thermodynamic neuron.
Virtual qubit	The two machine qubits define an effective transition with inverse temperature $\beta_v = (\beta_1\epsilon_1 - \beta_0\epsilon_0)/\epsilon_z$.	The input bath temperature is converted into an effective output bias.
Collector and modulator	Energy–conserving three–body swapping gives collector current; the modulator competes with it and fixes a stable output point.	The collector alone is only an amplifier; the modulator turns it into a gate with a threshold and noise margin.
Perceptron mapping	The general virtual temperature is a linear score in the input inverse temperatures, followed by a thermal sigmoid.	This explains why a single thermodynamic neuron realizes exactly linearly separable Boolean functions.
Non–Markovian extension	The Nakajima–Zwanzig memory kernel, influence functional, PT–MPO/TEMPO discretization, and convergence checks.	This shows what must replace the local Lindblad/reset picture when bath memory is important.
Floquet buffer	A periodically driven intermediate subsystem produces sidebands, spectral filtering, and stroboscopic control.	This is the proposed mechanism for reducing leakage and improving logical distinguishability.

2 The single qubit

A qubit is a two–level system with Hilbert space $\mathcal{H} = \mathbb{C}^2$, basis $\{|0\rangle, |1\rangle\}$, and free Hamiltonian

$$H = E_0 |0\rangle\langle 0| + E_1 |1\rangle\langle 1| \xrightarrow{E_0:=0} H = \epsilon n, \quad n := |1\rangle\langle 1|, \quad \epsilon := E_1 - E_0, \quad (1)$$

where only the gap ϵ is physical (the zero of energy is set at the ground state). In contact with a bath at inverse temperature $\beta = 1/k_B T$ the qubit relaxes to the Gibbs state

$$\tau(\beta) = \frac{e^{-\beta H}}{Z}, \quad Z = 1 + e^{-\beta\epsilon}, \quad \tau(\beta) = \frac{|0\rangle\langle 0| + e^{-\beta\epsilon} |1\rangle\langle 1|}{1 + e^{-\beta\epsilon}}. \quad (2)$$

The excited–state occupation is the Fermi–Dirac function, which is simultaneously the logistic sigmoid,

$$g(\beta\epsilon) := \langle 1 | \tau(\beta) | 1 \rangle = \frac{1}{1 + e^{\beta\epsilon}} = f(\beta\epsilon), \quad f(x) := \frac{1}{1 + e^x}. \quad (3)$$

This single identity — a thermal population is a sigmoid of an inverse temperature — is the seed of the entire neuron–perceptron correspondence developed in Section 6. The detailed–balance

ratio that every thermalization model must reproduce is

$$\frac{\langle 1 | \tau(\beta) | 1 \rangle}{\langle 0 | \tau(\beta) | 0 \rangle} = e^{-\beta\epsilon}. \quad (4)$$

3 Open quantum systems: where the bath Hamiltonian goes

This section resolves a point that is easy to miss in the working equations: if the qubit is coupled to a bath, *where is the qubit–bath interaction Hamiltonian, and why does it not appear explicitly in the master equation?* The answer is that it is present microscopically but, after the bath is eliminated, it is *converted* into the dissipator. We show this in the projection–operator (Nakajima–Zwanzig) language, which also exposes exactly where the Markovian description can fail — the entry point for Section 8.

3.1 A closed qubit cannot thermalize

Unitary evolution $\rho(t) = e^{-iHt}\rho(0)e^{iHt}$ of the qubit (1) generates only phases; the populations $\langle 0 | \rho | 0 \rangle, \langle 1 | \rho | 1 \rangle$ are constant. A closed qubit therefore cannot relax to (2). Dissipation, and hence computation–by–relaxation, is impossible without an environment: the bath is the only object that can change populations and drive the system to a steady state.

3.2 Microscopic model

Restore the environment as a bosonic reservoir bilinearly and transversally coupled to the qubit,

$$H_{\text{tot}} = \underbrace{H_S}_{\epsilon n} + \underbrace{H_B}_{\sum_k \omega_k a_k^\dagger a_k} + \underbrace{H_{SB}}_{\sigma_x \otimes \sum_k \lambda_k (a_k + a_k^\dagger)}. \quad (5)$$

The coupling is transverse ($\sigma_x = \sigma_+ + \sigma_-$) because the bath must be able to *flip* the qubit to exchange the quantum ϵ ; a longitudinal σ_z coupling would only dephase and never thermalize (a fact we exploit deliberately when the Floquet drive is chosen longitudinal in Section 9). The bilinearity is approximation (A2) in Table 1.

3.3 Nakajima–Zwanzig equation and the origin of memory

Write the exact Liouville–von Neumann equation $\dot{\rho}_{\text{tot}} = \mathcal{L}\rho_{\text{tot}}$, $\mathcal{L}(\cdot) = -i[H_{\text{tot}}, \cdot]$, and introduce the projection superoperators onto a reference bath state ρ_B ,

$$\mathcal{P}X = \text{Tr}_B(X) \otimes \rho_B, \quad \mathcal{Q} = 1 - \mathcal{P}. \quad (6)$$

Projecting the exact equation and formally solving the $\mathcal{Q}$ part gives the exact Nakajima–Zwanzig equation for the relevant part $\mathcal{P}\rho_{\text{tot}}$,

$$\frac{d}{dt}\mathcal{P}\rho_{\text{tot}}(t) = \mathcal{P}\mathcal{L}\mathcal{P}\rho_{\text{tot}}(t) + \int_0^t \mathcal{K}(t-\tau) \mathcal{P}\rho_{\text{tot}}(\tau) d\tau + \mathcal{I}(t), \quad (7)$$

$$\mathcal{K}(t-\tau) = \mathcal{P}\mathcal{L}\mathcal{Q}e^{\mathcal{Q}\mathcal{L}(t-\tau)}\mathcal{Q}\mathcal{L}\mathcal{P}, \quad \mathcal{I}(t) = \mathcal{P}\mathcal{L}\mathcal{Q}e^{\mathcal{Q}\mathcal{L}t}\mathcal{Q}\rho_{\text{tot}}(0). \quad (8)$$

For a factorized initial state $\mathcal{Q}\rho_{\text{tot}}(0) = 0$, so $\mathcal{I}(t) = 0$ (this is approximation (A1)). The memory kernel $\mathcal{K}$ is the mathematical origin of non–Markovianity: the present reduced motion depends on the system’s history. Equation (7) is exact; the Markovian master equation is what remains after three further approximations.

3.4 Born–Markov–secular reduction and the dissipator

Passing to the interaction picture, integrating once, tracing out the bath (using $\text{Tr}_B[B_a \rho_B] = 0$), and applying in turn the Born (A2), Markov (A3) and secular (A4) approximations yields the Gorini–Kossakowski–Sudarshan–Lindblad (GKSL) master equation

$$\dot{\rho}_S = -i[H_S + H_{\text{LS}}, \rho_S] + \sum_{\omega} \gamma(\omega) \left[A(\omega) \rho_S A^\dagger(\omega) - \frac{1}{2} \{A^\dagger(\omega) A(\omega), \rho_S\} \right], \quad (9)$$

with eigenoperators $A(\omega)$ of H_S (for the qubit, $A(\pm\epsilon) = \sigma_{\mp}$), rates $\gamma(\omega) = \int_{-\infty}^{\infty} e^{i\omega t} C(t) dt$ set by the bath correlation function $C(t) = \text{Tr}_B[\tilde{B}(t) B \rho_B]$, and a Lamb shift H_{LS} that merely renormalizes ϵ . For the qubit this is

$$\dot{\rho} = -i[H_S, \rho] + \gamma_{\downarrow}(\sigma_- \rho \sigma_+ - \frac{1}{2} \{\sigma_+ \sigma_-, \rho\}) + \gamma_{\uparrow}(\sigma_+ \rho \sigma_- - \frac{1}{2} \{\sigma_- \sigma_+, \rho\}). \quad (10)$$

This is the answer to the question posed above. The interaction Hamiltonian $H_{SB} = \sigma_x \otimes B$ no longer appears as a Hamiltonian: eliminating the bath has converted it into the two dissipative channels of (10) — $\gamma_{\downarrow}$ multiplying σ_- (emission), $\gamma_{\uparrow}$ multiplying σ_+ (absorption). The bath Hamiltonian H_B has disappeared entirely; the coupling survives only through the two rates. That is why the working equations carry a free Hamiltonian and a dissipator but no explicit qubit–bath interaction Hamiltonian.

The two rates are locked by the Kubo–Martin–Schwinger (detailed balance) condition $\gamma(-\omega) = e^{-\beta\omega} \gamma(\omega)$, so

$$\frac{\gamma_{\uparrow}}{\gamma_{\downarrow}} = e^{-\beta\epsilon}, \quad \Rightarrow \quad p_{\text{ss}} = \frac{\gamma_{\uparrow}}{\gamma_{\uparrow} + \gamma_{\downarrow}} = g(\beta\epsilon), \quad (11)$$

i.e. the qubit relaxes to exactly the thermal population (3).

3.5 The reset model

Since only populations matter for the logic, we adopt the simplest dissipator with the correct fixed point — the reset (collisional) model (A5),

$$\mathcal{L}^{(k)}[\rho] = \gamma_k (\text{Tr}_k[\rho] \otimes \tau(\beta_k) - \rho), \quad (12)$$

in which qubit k is, with rate γ_k , swapped for a fresh bath qubit in the thermal state $\tau(\beta_k)$. It manifestly satisfies local detailed balance $\mathcal{L}^{(k)}[\tau(\beta_k)] = 0$. The heat current into bath k is

$$j_k := \text{Tr}[H \mathcal{L}^{(k)}[\rho]] = \gamma_k \epsilon_k [g(\beta_k \epsilon_k) - p_k], \quad (13)$$

with p_k the excited population of qubit k . A qubit coupled to two baths acquires the sum of two dissipators, and the total current is the sum.

3.6 Multi–qubit machines, currents, entropy production

A machine of qubits $\{C_k\}$ has free Hamiltonian $H_0 = \sum_k \epsilon_k n_k$ and a time–independent, energy–preserving interaction H_{int} with $[H_{\text{int}}, H_0] = 0$ (A6): the interaction may only connect degenerate eigenstates of H_0 , shuffling excitations without changing the bare energy. Under weak coupling the machine obeys the local master equation (A7)

$$\dot{\rho} = -i[H_0 + H_{\text{int}}, \rho] + \sum_k \mathcal{L}^{(k)}[\rho]. \quad (14)$$

The second law for this class of machines is

$$\dot{\Sigma} := \dot{S}(\rho) - \sum_k \beta_k j_k \geq 0, \quad S(\rho) = -\text{Tr}[\rho \ln \rho], \quad (15)$$

and in a nonequilibrium steady state ($\dot{S} = 0$) all produced entropy is the weighted heat dumped into the baths, $\dot{\Sigma} = -\sum_k \beta_k j_k$. This $\dot{\Sigma}$ is the “dissipation” whose trade-off against accuracy we quantify in Section 5. The heat/work split that defines j_k is itself an approximation (A11) that becomes ambiguous at strong coupling — the caveat that Sections 8–13 take seriously.

4 The virtual qubit and the resonant interaction

4.1 Virtual qubit and virtual temperature

Take two qubits C_0, C_1 (gaps $\epsilon_0 \geq \epsilon_1$) each in their own bath at β_0, β_1 . Absent interaction the joint state is the product $\tau_{C_0}(\beta_0) \otimes \tau_{C_1}(\beta_1)$. The two crossed states

$$|0\rangle_v := |0\rangle_{C_0} |1\rangle_{C_1}, \quad |1\rangle_v := |1\rangle_{C_0} |0\rangle_{C_1} \quad (16)$$

span a *virtual qubit* of gap $\epsilon_v = \epsilon_0 - \epsilon_1$. Its population ratio defines a *virtual temperature* exactly as in (4):

$$e^{-\beta_v \epsilon_v} = \frac{P_{10}}{P_{01}} = \frac{e^{-\beta_0 \epsilon_0}}{e^{-\beta_1 \epsilon_1}} \Rightarrow \boxed{\beta_v = \frac{\epsilon_0}{\epsilon_0 - \epsilon_1} \beta_0 - \frac{\epsilon_1}{\epsilon_0 - \epsilon_1} \beta_1.} \quad (17)$$

The virtual inverse temperature is a *linear combination of the input inverse temperatures* — the perceptron weighted sum. It can lie outside $[\beta_0, \beta_1]$ and even go negative (population inversion), giving the refrigerator / heat-pump / heat-engine regimes as β_1 is swept at fixed β_0 .

4.2 The resonant three-body interaction

Add a target qubit C_z intended to inherit β_v . For a weak, energy-preserving coupling to transfer excitations resonantly between the virtual qubit and C_z , choose

$$\boxed{\epsilon_z = \epsilon_v = \epsilon_0 - \epsilon_1,} \quad (18)$$

which makes $|100\rangle$ (energy ϵ_0) and $|011\rangle$ (energy $\epsilon_1 + \epsilon_z$) degenerate. The unique Hermitian, energy-preserving operator connecting only this pair is the three-body swap

$$\boxed{H_{\text{int}} = \chi(|100\rangle\langle 011| + |011\rangle\langle 100|) = \chi(\sigma_+^{(0)}\sigma_-^{(1)}\sigma_-^{(z)} + \text{h.c.}),} \quad (19)$$

with ordering $(C_0 C_1 C_z)$. In virtual-qubit variables $H_{\text{int}} = \chi(|1\rangle_v\langle 0|_{C_z} + |0\rangle_v\langle 1|_{C_z} + \text{h.c.})$: a resonant swap between the virtual qubit and C_z . Five points fix its form.

- (i) *Energy conservation / RWA.* Under (18), $|100\rangle$ and $|011\rangle$ are the only degenerate pair differing by an allowed excitation rearrangement; every other off-diagonal term changes the bare energy and is dropped by the rotating-wave argument (A6).
- (ii) *Three-body, not pairwise.* The virtual qubit is a joint property of C_0, C_1 ; a two-body $\sigma_+^{(0)}\sigma_-^{(1)}$ only rotates the excitation between them and transfers nothing to C_z .

- (iii) *Physical action.* The swap places the virtual qubit in thermal contact with C_z , so in steady state C_z carries β_v .
- (iv) *Dropped partners.* Terms creating three excitations oscillate at $\epsilon_0 + \epsilon_1 + \epsilon_z$ and average away with error $O(\chi/\epsilon)$.
- (v) *No bare C_0 – C_1 or $\sigma_z\sigma_z$ coupling.* These would hybridize the inputs and destroy the product–thermal structure from which β_v was defined.

Remark 4.1. The resonant pair is $|100\rangle \leftrightarrow |011\rangle$. The pair $|101\rangle \leftrightarrow |010\rangle$ is detuned by $\epsilon_0 + \epsilon_z - \epsilon_1 = 2\epsilon_z \neq 0$ and is dynamically inert at $\chi \ll \epsilon_z$; building H_{int} from it is a common but fatal coding error. This form is confirmed by the baseline paper’s interaction (its Eq. (14)).

5 The NOT gate: collector, modulator, dynamics, logic

5.1 The collector alone is a linear amplifier

The collector (C_0, C_1, C_z with $H_0 + H_{\text{int}}$ and reset baths, rates γ for C_0, C_1 and μ for C_z) delivers to the output bath B_z the current

$$j_C = \mu\epsilon_z[g_z(\beta_z) - g_z(\beta_v)], \quad g_z(x) := g(x\epsilon_z). \quad (20)$$

In isolation $j_C = 0$ gives $\beta_z = \beta_v$, an *affine* function of the input by (17): an inverting linear amplifier, which amplifies input noise together with the signal.

5.2 Why the modulator is needed

A good gate must *saturate*: flat inside a logical region (noise rejected), steep across the decision boundary. This nonlinearity is supplied by the modulator $\mathcal{M}$ — a single qubit $H_{\mathcal{M}} = \epsilon_z n_{\mathcal{M}}$ in simultaneous contact with a fixed reference bath B_r (rate γ) and the output bath B_z (rate μ'). Its excited population obeys $\dot{p}_{\mathcal{M}} = \gamma[g_z(\beta_r) - p_{\mathcal{M}}] + \mu'[g_z(\beta_z) - p_{\mathcal{M}}]$, with steady state

$$p_{\mathcal{M}}^{\text{ss}} = \frac{\gamma g_z(\beta_r) + \mu' g_z(\beta_z)}{\gamma + \mu'} \xrightarrow{\mu' \ll \gamma} g_z(\beta_r) \quad (\text{A8}), \quad (21)$$

so the modulator is a heat valve delivering

$$j_{\mathcal{M}} = \mu'\epsilon_z[g_z(\beta_z) - g_z(\beta_r)] \quad (22)$$

that continuously pulls β_z toward β_r , competing with the collector. Without the second bath there is no valve and the gate degenerates to the linear amplifier: the modulator is what turns the weighted sum into a saturating (sigmoidal) activation.

5.3 Dynamics, fixed point, transfer characteristic

The finite output reservoir (heat capacity $\mathcal{C}$) evolves by the calorimetric law

$$\dot{\beta}_z = -\frac{\beta_z^2}{\mathcal{C}}(j_C + j_{\mathcal{M}}), \quad (23)$$

consistent with inserting the steady currents (20),(22) because $\gamma \gg \mu, \mu'$ (adiabatic elimination, A9). The fixed point $j_C + j_M = 0$ gives, with $\Delta := \mu/(\mu + \mu')$,

$$g_z(\beta_z^\infty) = \Delta g_z(\beta_v) + (1 - \Delta) g_z(\beta_r), \quad \beta_z^\infty = \frac{1}{\epsilon_z} \ln \left[\frac{1 - q^*}{q^*} \right], \quad q^* = g_z(\beta_z^\infty). \quad (24)$$

Fixing the modulator so the output is confined to $[\beta_{\text{hot}}, \beta_{\text{cold}}]$ requires $\Delta = g_z(\beta_{\text{hot}}) - g_z(\beta_{\text{cold}})$ and $g_z(\beta_r) = g_z(\beta_{\text{cold}})/(1 - \Delta)$, giving the exact transfer characteristic

$$\beta_z^\infty = \frac{1}{\epsilon_z} \ln [Q(\beta_v)^{-1} - 1], \quad Q(\beta_v) = g_z(\beta_{\text{hot}})g_z(\beta_v) + g_z(\beta_{\text{cold}})(1 - g_z(\beta_v)). \quad (25)$$

5.4 Emergence of the sigmoid

Because $\beta_v \epsilon_z = \beta_0 \epsilon_0 - \beta_1 \epsilon_1$ exactly, $g_z(\beta_v) = f(\beta_0 \epsilon_0 - \beta_1 \epsilon_1)$, and expanding (25) for small ϵ_z (A10) gives

$$\beta_z^\infty \simeq \beta_{\text{cold}} + (\beta_{\text{hot}} - \beta_{\text{cold}}) f(\beta_0 \epsilon_0 - \beta_1 \epsilon_1), \quad (26)$$

a sigmoidal interpolation between the logical rails whose argument is the perceptron pre-activation. The gain at threshold ($\beta_1^* \approx \beta_0$) is $\mathcal{A}_{\text{max}} = -\epsilon_1 \Delta \beta / 4$ with $\Delta \beta = \beta_{\text{cold}} - \beta_{\text{hot}}$: steeper gate $\Rightarrow$ larger $\epsilon_1 \Rightarrow$ more dissipation.

5.5 Logic encoding and the error–dissipation trade–off

The bit is encoded and decoded by

$$x = \begin{cases} 0, & \beta_1 = \beta_{\text{hot}} \\ 1, & \beta_1 = \beta_{\text{cold}} \end{cases} \quad y = \Theta(\beta_z^\infty - \beta_{\text{th}}), \quad \beta_{\text{th}} = \frac{1}{2}(\beta_{\text{hot}} + \beta_{\text{cold}}), \quad (27)$$

with logical noise margin $m_{\text{NOT}} = \min[\beta_z^\infty(\beta_{\text{hot}}) - \beta_{\text{th}}, \beta_{\text{th}} - \beta_z^\infty(\beta_{\text{cold}})]$. Finite-bath fluctuations ($\sim \mathcal{N}(\beta_z^\infty, \mathcal{C})$) give a nonzero decoding error $\langle \xi \rangle$; sharpening the gate (larger ϵ_1) lowers $\langle \xi \rangle$ but raises the integrated entropy production $\langle \Sigma \rangle$ of (15). This monotone accuracy–dissipation trade–off is the central thermodynamic cost of the gate and the baseline the Floquet buffer seeks to improve.

6 The general neuron: perceptron equivalence and networks

The n –input collector (C_0 reference, $C_1 \dots C_n$ inputs, C_z output) selects a virtual qubit via a binary vector $h = (h_0, \dots, h_n)$; its virtual temperature is

$$\beta_v = \frac{1}{\epsilon_z} \sum_{i=0}^n (-1)^{h_i} \beta_i \epsilon_i, \quad (28)$$

again a linear score of the inputs. With overall scale α , the output is $\beta_z^\infty = f(\alpha(w_0 + \sum_i w_i \beta_i)) + O(\epsilon_z)$ — exactly a sigmoid perceptron. Hence a single neuron realizes precisely the linearly–separable functions; its hardware is read off a trained perceptron by $h_k = \Theta(-w_k)$, $\epsilon_k = \alpha|w_k|$ (with the reference set by $|\epsilon_z - \sum_k w_k|$), $\beta_0 = |w_0|/|\epsilon_z - \sum_k w_k|$, and α trading accuracy against dissipation. NOR ($h = (0, 1, 1)$, $\epsilon = \alpha(\epsilon_z + 4, 2, 2)$) and 3–MAJORITY ($h = (1, 0, 0, 0)$, $\epsilon = \alpha(\epsilon_z + 12, 3, 3, 3)$) follow directly; non–separable functions such as XOR require a small network (NAND, OR $\rightarrow$ AND) whose neurons are composed through a clock–gated readout that makes each finite output bath act as an infinite input bath for the next neuron.

7 Baseline reconstruction: numerical reproduction

The analytic model of Sections 4–6 was reconstructed numerically to fix the Markovian reference against which every extension is judged. The reconstruction follows a fixed order: (i) the thermal occupation (3); (ii) the virtual temperature (17); (iii) the reset currents (20),(22); (iv) the output fixed point (24); (v) the threshold decoder (27); (vi) an energy–scale sweep. The closed–form fixed point $\beta_z^* = \epsilon_z^{-1} \ln[(1 - q^*)/q^*]$ with $q^* = [\mu g_z(\beta_v) + \mu' g_z(\beta_r)]/(\mu + \mu')$ lets truth tables be evaluated without time–integration error. The reconstructed logical noise margin is $m_{\text{NOT}} = \min[\beta_z^*(\beta_{\text{hot}}) - \beta_{\text{th}}, \beta_{\text{th}} - \beta_z^*(\beta_{\text{cold}})]$; a positive margin certifies the Boolean NOT.

Results at a glance

The work completed in this repository establishes four concrete outputs. First, the baseline autonomous NOT gate was reconstructed from the closed–form reset equations, giving the expected inversion of the input temperature into the output temperature and a positive noise margin in the working parameter window. Second, the energy–scale sweep reproduced the error–dissipation trade–off: sharper transfer curves reduce logical error but require larger entropy production. Third, the same perceptron construction was extended to NOR, MAJORITY, and a composed XOR network, confirming the linearly–separable limitation of one thermodynamic neuron and the need for networks for non–separable logic. Fourth, the non–Markovian and Floquet studies produced the comparison, convergence, ablation, and robustness figures used later in this part, which define the parameter regimes where reservoir memory, drive frequency, and buffer coupling either preserve or destroy useful logic.

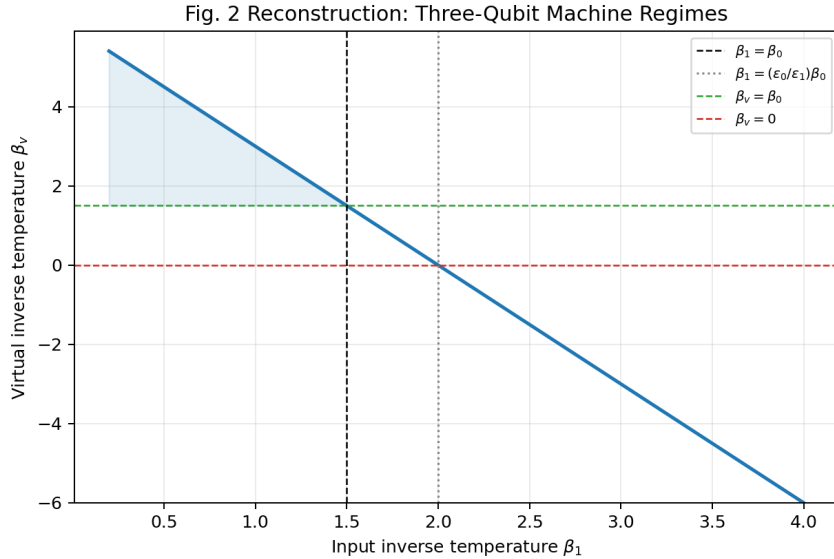


Figure 1: Reconstructed virtual–temperature regimes of the collector, validating the sign and scale of (17) before any decoding.

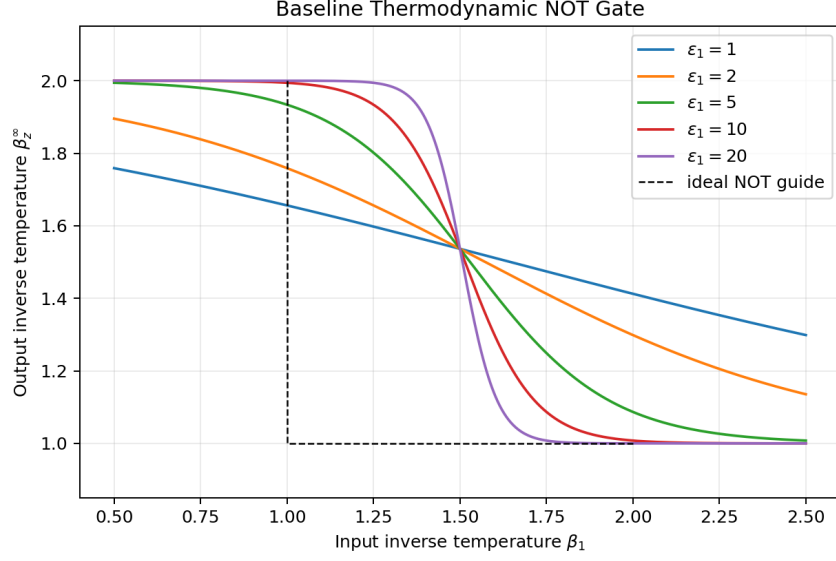


Figure 2: Reconstructed NOT transfer curves (25) for several collector energy scales. The gate is realized when the hot- and cold-input outputs lie on opposite sides of β_{th} ; increasing ϵ_1 sharpens the curve toward the ideal step.

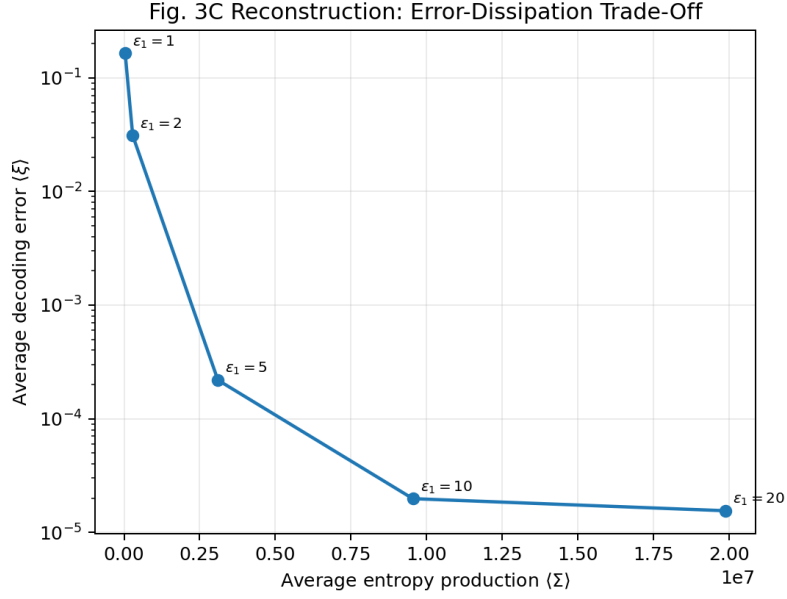


Figure 3: Reconstructed error-dissipation trade-off: lowering the decoding error requires larger entropy production, as anticipated in Section 5.5.

Table 2: Representative error–dissipation data for increasing collector energy scale ϵ_1 (uniform inputs).

ϵ_1	Average error	Average entropy production	Hot-input output
1	1.6477×10^{-1}	3.7266×10^4	1.6561
2	3.1114×10^{-2}	2.8675×10^5	1.7588
5	2.1882×10^{-4}	3.1029×10^6	1.9338
10	1.9770×10^{-5}	9.5496×10^6	1.9942
20	1.5480×10^{-5}	1.9882×10^7	2.0000

The same reconstruction reproduces multi-input and composed logic: the NOR response surface (Fig. 4), the three-input majority response (Fig. 5), and an XOR network assembled from thermodynamic–neuron subgates (Fig. 6), consistent with the perceptron correspondence of Section 6.

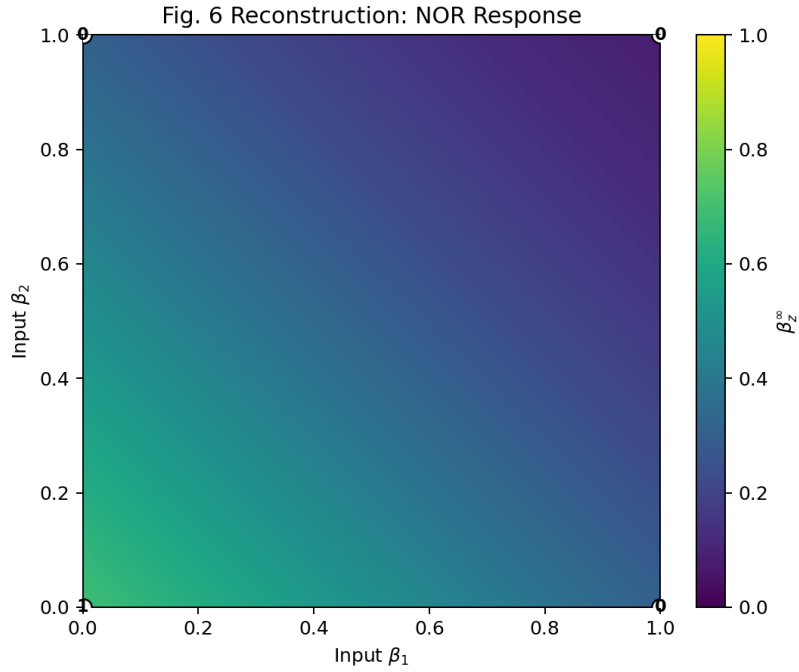


Figure 4: Reconstructed NOR response surface: high output only near the low–low input corner.

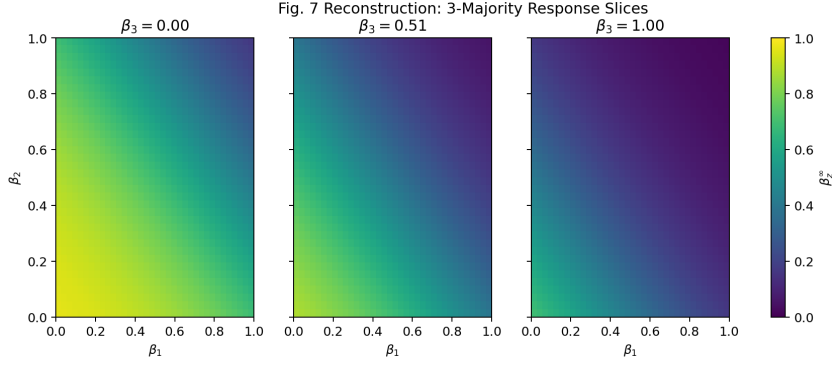


Figure 5: Reconstructed three-input majority response (two-dimensional slices), a linear thermodynamic score of the input temperatures.

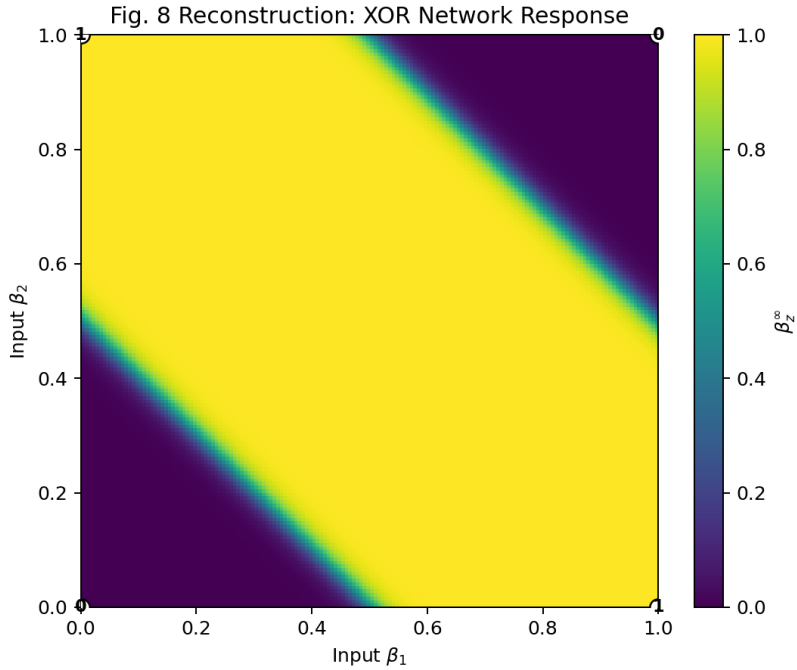


Figure 6: Reconstructed XOR response from a small network of neurons; XOR is not linearly separable and cannot be a single perceptron.

8 Beyond weak coupling: non-Markovian dynamics

The reset model rests on the whole approximation chain of Table 1. For a *structured* reservoir the Markov step (A3) is the first to fail: the memory kernel $\mathcal{K}$ of the exact Nakajima–Zwanzig equation (7) is no longer sharply peaked, and the reduced motion

$$\dot{\rho}_S(t) = -i[H_S, \rho_S(t)] + \int_0^t \mathcal{K}(t - \tau) \rho_S(\tau) d\tau \quad (29)$$

retains genuine history dependence: revivals, information backflow, and modified relaxation. A strong claim about non-Markovian thermodynamic logic therefore requires a numerically exact treatment of the memory, not a Lindblad surrogate.

8.1 Influence functional and process tensor

For a Gaussian bosonic bath linearly coupled to the system, the bath can be integrated out exactly. On a forward–backward path pair (s^+, s^-) the reduced propagator carries the Feynman–Vernon influence functional

$$\mathcal{F}[s^+, s^-] = \exp \left[- \int_0^t d\tau \int_0^\tau d\tau' (s_\tau^+ - s_\tau^-) (\eta(\tau - \tau') s_{\tau'}^+ - \eta^*(\tau - \tau') s_{\tau'}^-) \right], \quad (30)$$

with the bath response kernel set by the spectral density,

$$\eta(t) = \int_0^\infty d\omega J(\omega) \left[\coth\left(\frac{\beta\omega}{2}\right) \cos \omega t - i \sin \omega t \right], \quad J(\omega) = 2\alpha \omega^s e^{-\omega/\omega_c}. \quad (31)$$

Discretizing time into N slices makes the influence functional a multilinear object — the *process tensor* — mapping a sequence of control operations $\mathcal{A}_0, \dots, \mathcal{A}_{N-1}$ to the final state,

$$\rho_N = \Upsilon_{N:0} \star (\mathcal{A}_{N-1} \otimes \dots \otimes \mathcal{A}_0), \quad \Upsilon_{N:0} = \sum_{\{\alpha_k\}} M_{\alpha_{N-1}}^{[N]} \dots M_{\alpha_0}^{[1]} M^{[0]}. \quad (32)$$

8.2 MPS, MPO, TTN and why memory is compressible

Written as a matrix product operator along the time axis, the process tensor has bond dimension χ measuring the memory carried from past to future:

$$\Upsilon = \sum_{\{i_k, j_k\}} W^{[0]i_0 j_0} \dots W^{[N]i_N j_N} |i_0 \dots i_N\rangle \langle j_0 \dots j_N|. \quad (33)$$

A Markovian bath has $\chi = 1$; a non–Markovian but compressible bath has $1 < \chi \ll d^{2N}$, so singular–value truncation keeps the physically relevant temporal correlations without the exponential blow–up of an explicit system–plus–bath simulation (dimension $d_S d^N$). A tree–tensor–network (TTN) generalization places each reservoir of the neuron on a separate branch, which is why the numerical programme lists PT–MPO/TEMPO as the primary structured–bath backend and TTN as the fallback when memory growth dominates. This is the same chain / MPS machinery treated rigorously in Part II.

8.3 A single worked example: from the memory kernel to output logic

Consider one logical qubit coupled to a structured bath through a driven buffer, $S \leftrightarrow F(t) \leftrightarrow B$.

- (1) The full state obeys $\dot{\rho}_{SFB} = -i[H_{SFB}(t), \rho_{SFB}]$ and the logical state is $\rho_S = \text{Tr}_{FB} \rho_{SFB}$.
- (2) Tracing out buffer and bath gives a driven memory kernel $\mathcal{K}_F(t, \tau)$; tuning the drive (A, Ω) reshapes $\mathcal{K}_F$ and hence how information leaves or returns to the qubit.
- (3) Rather than construct $\mathcal{K}_F$ explicitly, the process–tensor method builds $\Upsilon_{n:0}$ and contracts it with the system operations (32).
- (4) For a binary gate, evolve two inputs $\rho_S^{(0)}(0), \rho_S^{(1)}(0)$ and read the distinguishability

$$D_{\text{out}} = \frac{1}{2} \|\rho_S^{(0)}(t_f) - \rho_S^{(1)}(t_f)\|_1. \quad (34)$$

The buffer is useful only if it improves D_{out} (or the decoding margin) at an acceptable drive cost.

8.4 Structured-bath benchmark and convergence

Before any gate claim, the solver must pass numerical-health checks: trace preservation $\delta_{\text{tr}} = |\text{Tr } \rho - 1|$, Hermiticity $\delta_{\text{herm}} = \|\rho - \rho^\dagger\|_F$, and convergence under Δt and memory time τ_{mem} . A single qubit in the bath (31) was tracked across sub-Ohmic, Ohmic, and super-Ohmic regimes (Table 3, Fig. 7); the largest trace deviation was 1.884×10^{-5} and the largest Hermiticity error 5.011×10^{-7} , with working point $(\Delta t, \tau_{\text{mem}}, \chi_{\text{svd}}) = (0.05, 1.0, 10^{-6})$. A first quick scan failed the 10^{-3} criterion (4.234×10^{-3}); refining Δt at fixed $\tau_{\text{mem}} = 1.0$ reduced the drift to 6.579×10^{-4} (pass).

Table 3: Single-qubit structured-bath benchmark across spectral regimes.

Regime	s	Final purity	Final σ_z	Runtime (s)
Sub-Ohmic	0.5	0.5070	-0.1184	20.996
Ohmic	1.0	0.5248	0.2230	6.582
Super-Ohmic	3.0	0.8421	0.8272	4.479

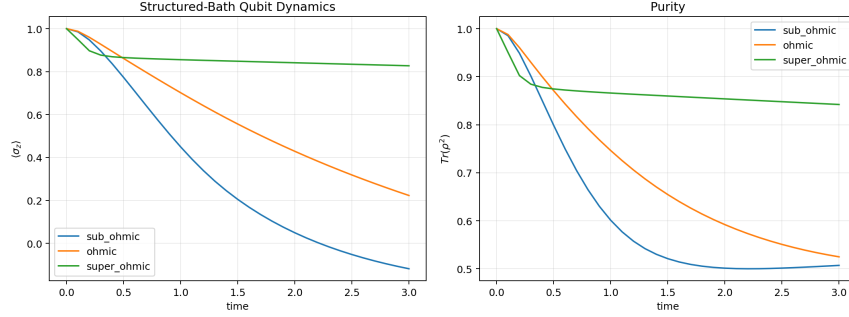


Figure 7: Reduced single-qubit dynamics for three structured-bath regimes — the numerical bridge between the Markovian neuron and later memory-bearing gate simulations.

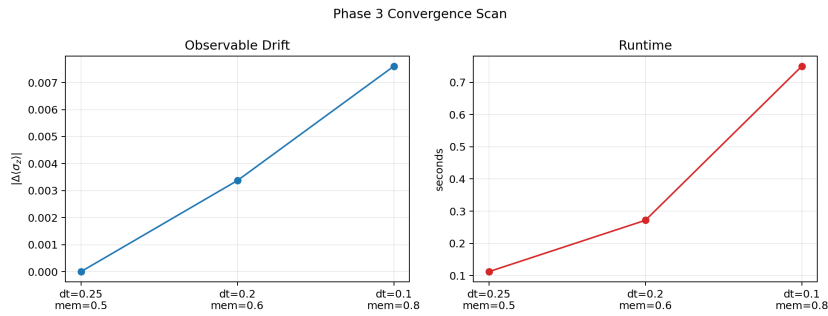


Figure 8: Quick convergence scan (structured-bath benchmark), used as a deliberate failure mode before interpreting non-Markovian logic.

9 The Floquet buffer

To reduce the logical qubit's direct exposure to the reservoir, a periodically driven buffer is inserted between subsystem and bath,

$$S \longleftrightarrow F(t) \longleftrightarrow B \quad \text{versus the direct} \quad S \longleftrightarrow B. \quad (35)$$

A minimal buffer Hamiltonian, as implemented in the bridge code, is

$$H_F(t) = \frac{\epsilon_F}{2} \sigma_z + A \cos(\Omega t) \sigma_x, \quad H_{SF} = g_{SF} \sigma_x^{(S)} \otimes \sigma_x^{(F)}, \quad T = \frac{2\pi}{\Omega}, \quad (36)$$

so the full bridge Hamiltonian is $H(t) = H_S + H_F(t) + H_{SF} + H_B + H_{FB}$. (The refined, purely longitudinal drive that yields clean Bessel sidebands is analyzed rigorously in Part II; here we keep the minimal implemented form.) In a weak Markovian treatment of the buffer–bath contact the logical qubit sees a time–periodic generator $\dot{\rho}_S = \mathcal{L}_{\text{eff}}(t)\rho_S$, $\mathcal{L}_{\text{eff}}(t+T) = \mathcal{L}_{\text{eff}}(t)$, whose rates sample the bath at drive–shifted frequencies $\omega + n\Omega$. This is the origin of Floquet filtering: the buffer presents the effective spectral density

$$J_{\text{eff}}(\omega) = \sum_n |c_n(A, \Omega)|^2 J(\omega + n\Omega), \quad (37)$$

whose weights $|c_n|^2$ are set by the drive.

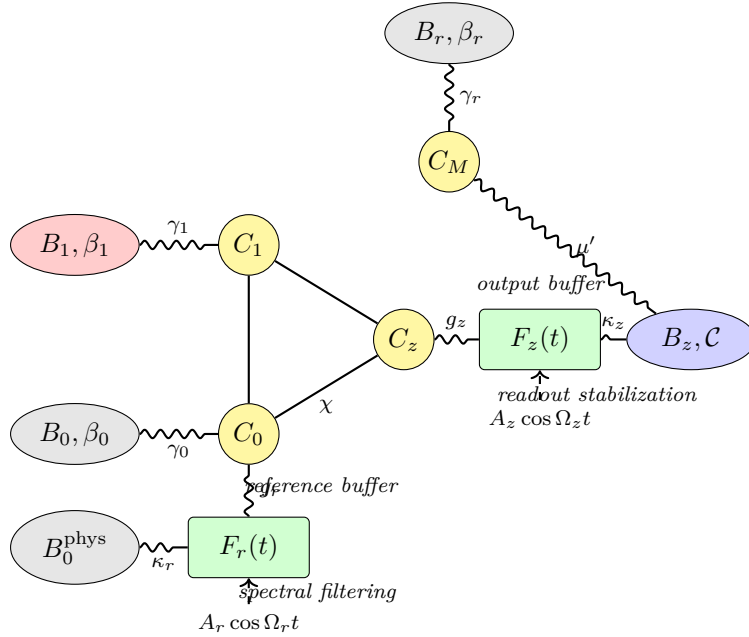


Figure 9: Floquet–buffered thermodynamic neuron architecture. The collector still forms the virtual temperature through the coherent three–body interaction among C_0, C_1, C_z , while driven intermediate systems $F_r(t)$ and $F_z(t)$ regulate the reference and output reservoir contacts. The direct autonomous contact is recovered by setting the buffer couplings to zero and reconnecting the corresponding bath directly to the qubit.

9.1 Operational metrics

The buffer is assessed operationally by the output trace distance (34) and by the drive–work proxy

$$W_{\text{proxy}} = \int_0^{t_f} \text{Tr}[\rho(t) \dot{H}_F(t)] dt = -A\Omega \int_0^{t_f} \sin(\Omega t) \langle \sigma_x \rangle_t dt. \quad (38)$$

This is an operational indicator, not a full strong-coupling thermodynamic audit (which must also track interaction and bath energy, cf. approximation (A11)).

9.2 Screening and ablation

A single run is insufficient evidence. Screening over drive amplitude, drive frequency and system–buffer coupling found positive trace–distance gain $\Delta D = D_{\text{buffer}} - D_{\text{direct}}$ in 14 of 15 configurations, with best gain 0.711315 at weak system–buffer coupling. The ablation (Table 4) isolates the drive: with $A = 0$ the buffer is a passive spacer and *degrades* distinguishability ($\Delta D = -0.149461$); the benefit appears only under periodic modulation, consistent with the sideband picture (37). The strongest gain at weak g_{SF} indicates the buffer acts as a selective mediated channel rather than a strongly hybridized extra degree of freedom.

Table 4: Ablation of the Floquet–buffer bridge. Direct baseline $D_{\text{direct}}(t_f) = 0.236928$; gain $\Delta D = D_{\text{buffer}} - D_{\text{direct}}$.

Case	$D_{\text{buffer}}(t_f)$	ΔD	Relative gain	$ W_{\text{proxy}} $
Full driven buffer	0.320480	0.083551	0.352644	0.176530
No periodic drive	0.087467	−0.149461	−0.630829	0
Weak drive	0.372211	0.135282	0.570984	0.086262
Strong drive	0.401688	0.164760	0.695401	0.087613
Weak system–buffer coupling	0.948243	0.711315	3.002237	0.019354
Strong system–buffer coupling	0.601144	0.364215	1.537240	0.439431
Weak buffer–bath contact	0.237160	0.000232	0.000979	0.021202
Strong buffer–bath contact	0.405590	0.168662	0.711869	0.398363

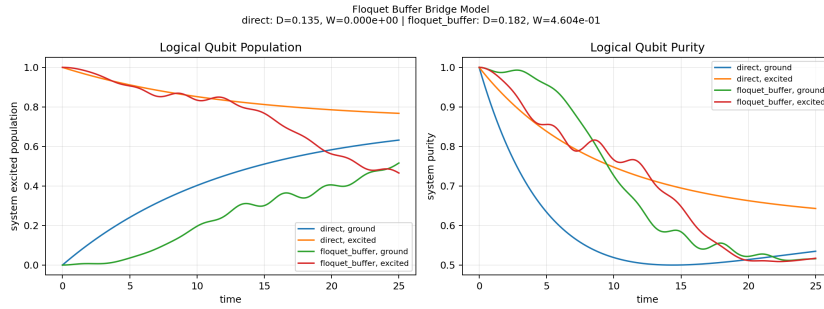


Figure 10: Direct versus Floquet–buffered dynamics in the bridge model.

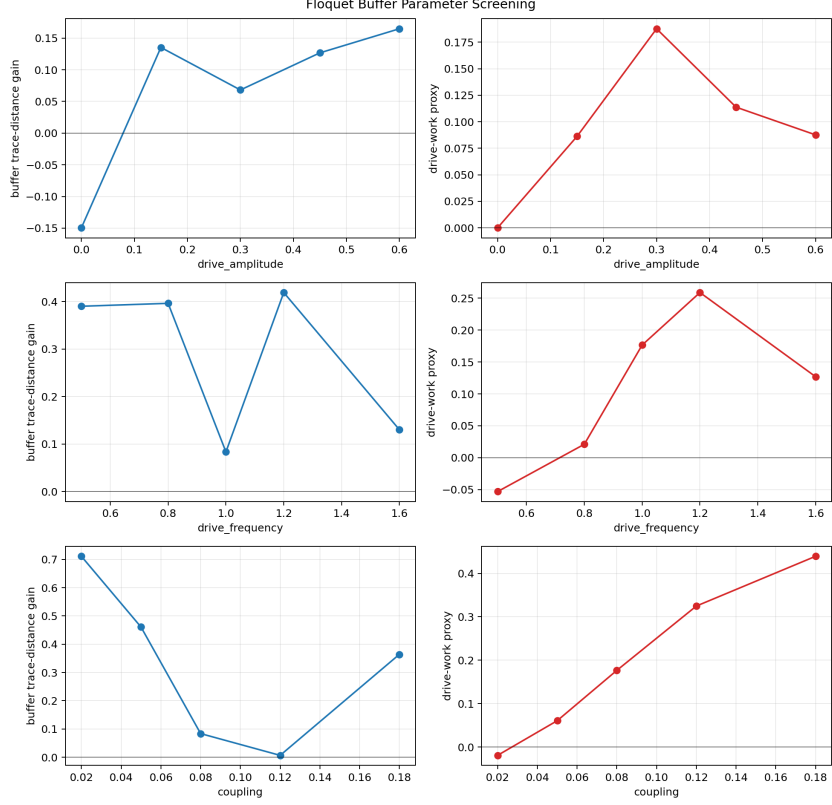


Figure 11: Floquet parameter sweep; best screened trace–distance gain 0.711315.

Embedding the buffer in the full three–qubit NOT surrogate preserves the truth table: both direct and buffered GKSL surrogates give $\mathcal{A}_{\text{NOT}} = 1$ with small drive–work proxy (Table 5). Under the Phase V PT–MPO/TEMPO structured bath the calibrated decoder (architecture–specific threshold $\beta_{\text{th}}^{(a)} = \frac{1}{2}[\beta_{\text{eff}}^{(a)}(0) + \beta_{\text{eff}}^{(a)}(1)]$) also passes, but the buffered margin is small ($\sim 2.6 \times 10^{-5}$ versus a direct margin $\sim 1.43 \times 10^{-1}$) — truth–table passing is binary, robustness is a margin statement.

Table 5: Main numerical outcomes across the staged programme. $\mathcal{A}_{\text{NOT}}$ is truth–table accuracy; $D(t_f)$ the final logical trace distance.

Experiment	Metric	Result	Interpretation
Floquet bridge	ΔD	0.0836 / 0.711 best	driven buffer can improve distinguishability
No–drive ablation	ΔD	−0.1495	passive buffer degrades; drive essential
3–qubit GKSL direct	$\mathcal{A}_{\text{NOT}}, D$	1.000, 0.0249	direct NOT passes
3–qubit GKSL buffered	$\mathcal{A}_{\text{NOT}}, D$	1.000, 0.0241	buffered NOT passes
PT–MPO/TEMPO direct	$\mathcal{A}_{\text{NOT}}, \text{margin}$	1.000, 1.43×10^{-1}	structured–bath direct passes
PT–MPO/TEMPO buffered	$\mathcal{A}_{\text{NOT}}, \text{margin}$	1.000, 2.60×10^{-5}	passes, close to boundary
GKSL phase map	robust points	68/80	broad passing region

10 Full GKSL, HEOM, and tensor–network gate validation

The final physics claim cannot rest on the bridge model alone. The complete validation stack should solve the same gate at three levels of description and demand that the logical truth table and the buffer advantage survive refinement.

10.1 Level 1: full time–dependent GKSL gate

The Markovian full-gate baseline keeps the collector, modulator, and buffer degrees of freedom explicitly in the system Hilbert space,

$$\rho_{\text{sys}}(t) \in \mathcal{B}(\mathcal{H}_{C_0} \otimes \mathcal{H}_{C_1} \otimes \mathcal{H}_{C_z} \otimes \mathcal{H}_{C_M} \otimes \mathcal{H}_{F_r} \otimes \mathcal{H}_{F_z}), \quad (39)$$

and evolves

$$\dot{\rho}_{\text{sys}}(t) = -i[H_{\text{gate}}(t), \rho_{\text{sys}}(t)] + \sum_{\alpha} \sum_{\omega} \gamma_{\alpha}(\omega) \left[L_{\alpha, \omega} \rho_{\text{sys}} L_{\alpha, \omega}^{\dagger} - \frac{1}{2} \{L_{\alpha, \omega}^{\dagger} L_{\alpha, \omega}, \rho_{\text{sys}}\} \right], \quad (40)$$

with $H_{\text{gate}}(t)$ containing the resonant collector coupling χ , the modulator branch, and the driven buffer Hamiltonians $H_{F_r}(t)$ and $H_{F_z}(t)$. The direct and buffered gates are compared using identical input temperatures and identical decoding:

$$\mathcal{A}_{\text{NOT}} = \frac{1}{2} \sum_{x \in \{0,1\}} \mathbf{1}[\Theta(\beta_z^{\infty}(x) - \beta_{\text{th}}) = 1 - x], \quad m = \min_x |\beta_z^{\infty}(x) - \beta_{\text{th}}|. \quad (41)$$

The GKSL level is fast enough for sweeps over drive amplitude A , drive frequency Ω , system–buffer coupling g_{SF} , and bath-contact scale κ . Its job is to find candidate operating regions, not to prove strong-coupling physics.

10.2 Level 2: HEOM benchmark for structured reservoirs

To include reservoir memory without replacing it by a local dissipator, each structured bath is represented through a correlation-function decomposition

$$C_{\alpha}(t) = \int_0^{\infty} J_{\alpha}(\omega) \left[\coth\left(\frac{\beta_{\alpha}\omega}{2}\right) \cos \omega t - i \sin \omega t \right] d\omega \simeq \sum_{k=0}^{K_{\alpha}} c_{\alpha k} e^{-\nu_{\alpha k} t}. \quad (42)$$

For a Drude–Lorentz bath, $J_{\alpha}(\omega) = 2\lambda_{\alpha}\gamma_{\alpha}\omega/(\omega^2 + \gamma_{\alpha}^2)$, the first exponential has $\nu_{\alpha 0} = \gamma_{\alpha}$ and the remaining terms are Matsubara or Padé frequencies. The HEOM propagates the physical density matrix ρ_0 together with auxiliary density operators (ADOs) $\rho_{\mathbf{n}}$:

$$\begin{aligned} \dot{\rho}_{\mathbf{n}} = & -i[H_{\text{gate}}(t), \rho_{\mathbf{n}}] - \sum_{\alpha k} n_{\alpha k} \nu_{\alpha k} \rho_{\mathbf{n}} - i \sum_{\alpha} [V_{\alpha}, \rho_{\mathbf{n}+\mathbf{e}_{\alpha k}}] \\ & - i \sum_{\alpha k} n_{\alpha k} (c_{\alpha k} V_{\alpha} \rho_{\mathbf{n}-\mathbf{e}_{\alpha k}} - c_{\alpha k}^* \rho_{\mathbf{n}-\mathbf{e}_{\alpha k}} V_{\alpha}), \end{aligned} \quad (43)$$

where V_{α} is the system operator coupled to bath α . The hierarchy is truncated by tier $N_{\text{tier}} = \sum_{\alpha k} n_{\alpha k}$ and by the number K_{α} of exponential terms. The validation checks are

$$\Delta_{\text{HEOM}} = \max_t \left\| \rho_0^{(N_{\text{tier}}, K)}(t) - \rho_0^{(N_{\text{tier}}+1, K+1)}(t) \right\|_1, \quad \Delta_{\text{GKSL/HEOM}} = \max_t \left\| \rho_{\text{GKSL}}(t) - \rho_0(t) \right\|_1. \quad (44)$$

HEOM is the preferred benchmark for reduced output or output-buffer subsystems because it is numerically controlled and keeps the bath memory kernel explicitly. For the full six-qubit-plus-buffer gate it may become large, so it should first validate the reference/output branches and then selected full-gate points from the GKSL phase map.

Computed HEOM output-buffer benchmark. The first HEOM benchmark was run for a structured Drude–Lorentz output environment with hierarchy depth $N_{\text{tier}} = 4$, convergence checked against depth 5, two Matsubara terms, and a driven qubit buffer. The direct structured-bath branch gives final logical trace distance $D_{\text{direct}} = 0.417302$, while the Floquet-buffered branch gives $D_{\text{buffer}} = 0.799184$. Thus the HEOM trace-distance gain at this working point is

$$\Delta D_{\text{HEOM}} = D_{\text{buffer}} - D_{\text{direct}} = 0.381882. \quad (45)$$

The maximum depth-refinement trace-distance drift is 5.111×10^{-7} for the direct branch and 1.198×10^{-6} for the buffered branch, so the result is converged at the plotted precision. The result should be read as an output-stage HEOM validation, not yet as the full six-qubit gate HEOM calculation.

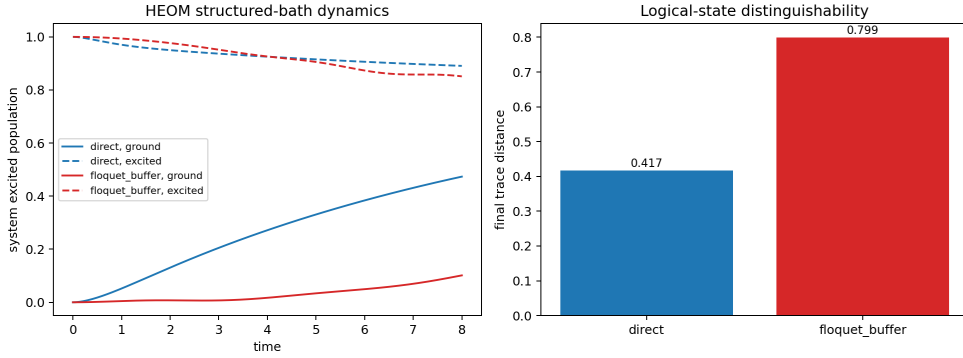


Figure 12: HEOM structured-bath validation of the output stage. The Floquet-buffered branch preserves a larger final distinguishability between the two logical initial states than the direct structured-bath contact at the same bath parameters.

HEOM depth scaling and exact-state MPS representability. The HEOM hierarchy was also swept over depths $N_{\text{tier}} = 2, 3, 4, 5$ on a shortened scaling run. At depth 4 the direct structured subsystem has $D = 0.522348$ with depth-5 drift 5.62×10^{-8} , while the Floquet-buffered subsystem has $D = 0.855945$ with drift 4.46×10^{-7} . Thus the HEOM result is already stable by depths 3–4 for this benchmark.

For the preliminary tensor-network estimate, each reservoir was replaced by a finite spin chain. The state was propagated exactly and the bond dimension needed to represent it as an MPS was inferred from its Schmidt spectra; this was not an MPS time evolution. For bath-chain lengths $L = 2, \dots, 6$, the direct subsystem required maximum bond dimension $\chi = 2, 4, 4, 6, 6$, while the Floquet-buffered chain required $\chi = 4, 4, 6, 6, 6$. The buffered geometry therefore costs an extra local site and a slightly larger small- L bond, but it does not show runaway bond growth in this benchmark. At $L = 6$ the direct chain gives $D = 0.748064$ and the buffered chain gives $D = 0.791114$.

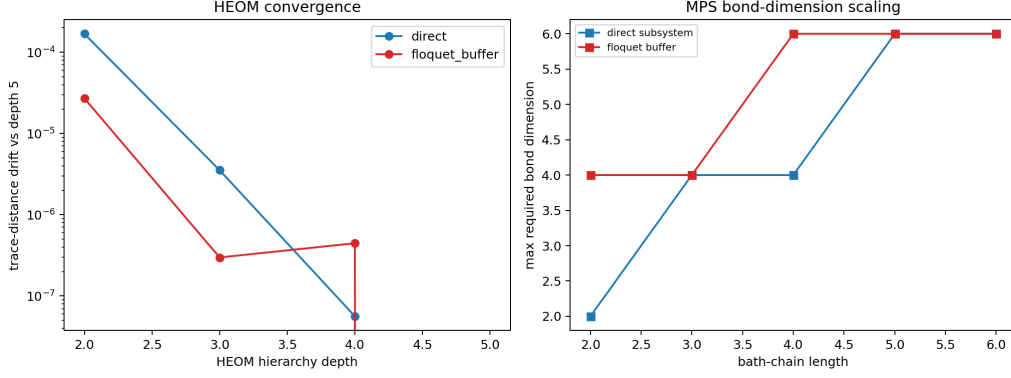


Figure 13: Scaling validation. Left: HEOM convergence versus hierarchy depth, measured by trace-distance drift relative to depth 5. Right: exact-state MPS representability versus structured-bath chain length for the direct subsystem and the Floquet-buffered subsystem.

Energy/work comparison and practical interpretation. The direct structured-bath subsystem uses no external drive work: $W_{\text{drive}} = 0$. It is therefore the lower-energy device if one counts only externally supplied work. Its drawback is weaker logical separation: in the HEOM depth-4 scaling run $D_{\text{direct}} = 0.522348$. The Floquet-buffered subsystem consumes clock/drive work $W_{\text{net}} = -0.406107$, $W_{+} = 0.038648$, and $W_{\text{abs}} = 0.483403$, and has $D_{\text{buffer}} = 0.855945$. The conservative work-throughput-per-distinguishability proxy is therefore $W_{\text{abs}}/D_{\text{buffer}} = 0.564759$. Thus the answer to “which uses less energy?” is: the direct subsystem uses less external energy, but the buffered subsystem buys stronger isolation and readout distinguishability by spending drive work. The useful engineering question is therefore not absolute work alone, but whether the improved noise margin is worth the supplied AC power.

Practically, the architecture can be implemented in platforms that allow tunable qubit gaps and engineered reservoirs. In superconducting circuits, C_z and F_z can be transmon-like qubits or a qubit plus resonator, the buffer gap can be modulated by flux or microwave driving, and the structured bath can be an impedance-engineered transmission line or a lossy resonator chain. In trapped ions or quantum dots, the same logic is implemented by a driven auxiliary level coupled to phonon/electronic reservoir modes. The experimental signature is simple: prepare the two logical input states, run the direct and buffered devices to the same readout time, measure the output populations or trace-distance proxy, and compare the gain against calibrated drive power.

Broad HEOM Floquet phase sweep. To move beyond a single hand-picked point, the HEOM output-stage model was swept over drive amplitude $A \in \{0, 0.2, 0.35, 0.55\}$, drive frequency $\Omega \in \{0.8, 1.2, 1.8\}$, system-buffer coupling $g_{SF} \in \{0.04, 0.08, 0.14\}$, reorganization energy $\lambda \in \{0.025, 0.045, 0.075\}$, and bath cutoff $\gamma_c \in \{0.8, 1.8, 3.0\}$, giving 324 buffered HEOM points at hierarchy depth 3. Of these, 284 points improve the direct structured-bath distinguishability and 229 points satisfy the robust criterion $D_{\text{buffer}} \geq 0.75$ with positive gain. The best absolute-gain point is

$$A = 0.55, \quad \Omega = 1.2, \quad g_{SF} = 0.04, \quad \lambda = 0.075, \quad \gamma_c = 0.8,$$

with $D_{\text{direct}} = 0.353784$, $D_{\text{buffer}} = 0.977156$, and $\Delta D = 0.623372$. The drive-power sign changes during the cycle, so three distinct work quantities are retained:

$$W_{\text{net}} = \int_0^{t_f} P(t) dt, \quad W_+ = \int_0^{t_f} \max[P(t), 0] dt, \quad W_{\text{abs}} = \int_0^{t_f} |P(t)| dt. \quad (46)$$

Here W_{net} permits ideal work recovery, W_+ counts energy supplied to the quantum system, and W_{abs} is the conservative cycle-throughput proxy when neither sign of power is assumed free. At the best-gain point these are $W_{\text{net}} = -0.539063$, $W_+ = 0.010179$, and $W_{\text{abs}} = 0.559420$, giving $\Delta D/W_{\text{abs}} = 1.114318$. This correction matters: taking only $|W_{\text{net}}|$ can produce a false near-zero cost through cancellation. The best driven point under the conservative ratio is $A = 0.2, \Omega = 0.8, g_{SF} = 0.04, \lambda = 0.075, \gamma_c = 0.8$, with $D_{\text{buffer}} = 0.968718$, $\Delta D = 0.614933$, $W_{\text{abs}} = 0.090825$, and $\Delta D/W_{\text{abs}} = 6.770543$.

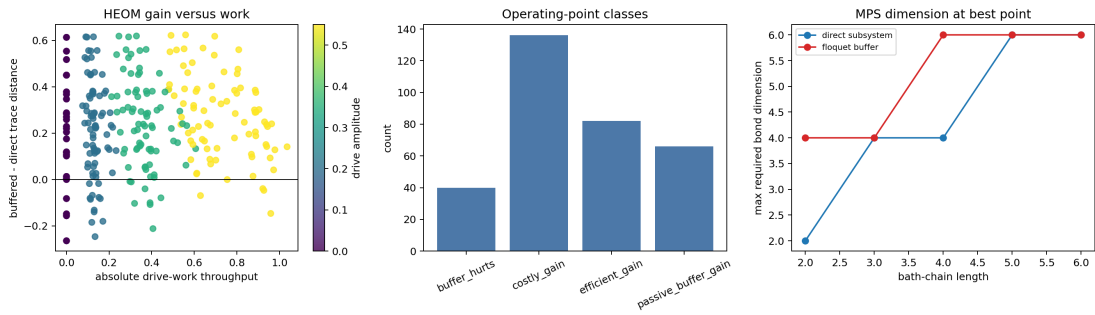


Figure 14: Broad HEOM phase-sweep summary. The sweep compares output-stage direct structured-bath contact against Floquet-buffered contact over drive, coupling, and bath-memory parameters; the MPS panel shows the bond-dimension estimate at the best-gain operating point.

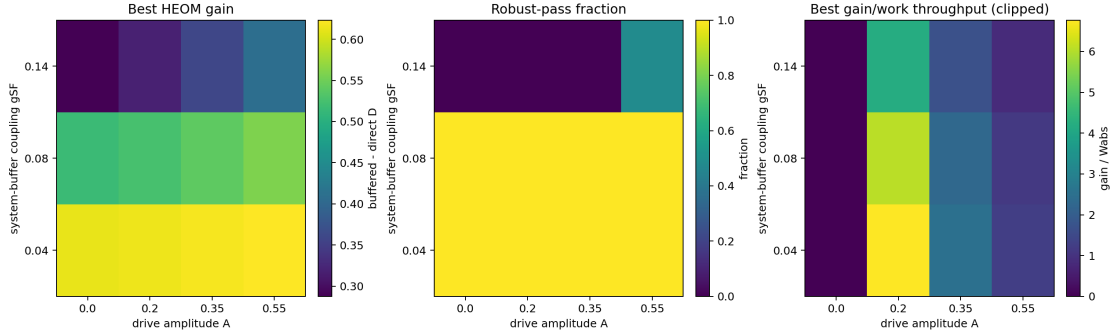


Figure 15: HEOM operating regions compressed over bath parameters and drive frequency. The panels show best gain, robust-pass fraction, and clipped gain/work over drive amplitude and system-buffer coupling.

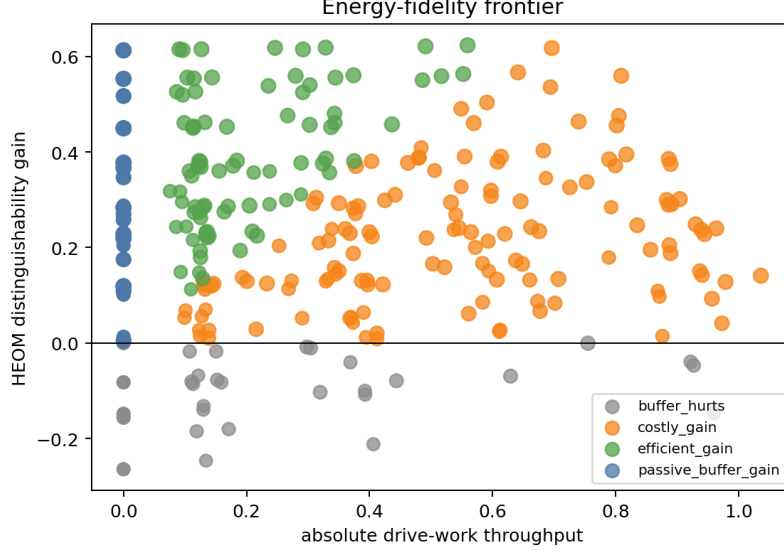


Figure 16: Energy–fidelity frontier from the HEOM phase sweep. Direct operation has zero external drive work; buffered operation spends absolute drive-work throughput to gain output distinguishability. Marker size scales with buffered final distinguishability.

Six-state HEOM channel tomography. A two-state truth-table test probes only one diameter of the Bloch sphere. For a genuine channel test we propagate the six Pauli eigenstates $\rho_{\pm j} = (I \pm \sigma_j)/2$ for $j = x, y, z$. Linearity then reconstructs the action of the channel $\mathcal{E}$ without an ancilla:

$$\mathcal{E}(I) = \mathcal{E}(\rho_{+z}) + \mathcal{E}(\rho_{-z}), \quad \mathcal{E}(\sigma_j) = \mathcal{E}(\rho_{+j}) - \mathcal{E}(\rho_{-j}). \quad (47)$$

The Pauli-transfer matrix and normalized Choi matrix follow as

$$R_{\mu\nu} = \frac{1}{2} \text{Tr}[\sigma_\mu \mathcal{E}(\sigma_\nu)], \quad J_{\mathcal{E}} = \frac{1}{4} \sum_{\nu=0}^3 \mathcal{E}(\sigma_\nu) \otimes \sigma_\nu^T. \quad (48)$$

Trace preservation demands $R_{0\nu} = \delta_{0\nu}$ and complete positivity demands $J_{\mathcal{E}} \succeq 0$. If $M = (R_{ij})_{i,j=1}^3$ is the Bloch block, the directional retention, unitarity, and volume contraction are

$$D_j = \frac{1}{2} \|\mathcal{E}(\rho_{+j}) - \mathcal{E}(\rho_{-j})\|_1, \quad u = \frac{1}{3} \text{Tr}(M^T M), \quad V = |\det M|. \quad (49)$$

For a target unitary with Bloch rotation O_U , the average fidelity is

$$F_{\text{avg}}(U, \mathcal{E}) = \frac{3 + \text{Tr}(O_U^T M)}{6}. \quad (50)$$

Maximizing this expression over $O_U \in SO(3)$ by the singular-value decomposition of M separates a correctable coherent frame rotation from irreversible contraction.

At the best-gain point, the direct channel gives $D_{\min} = 0.353784$, $u = 0.224019$, $V = 0.092394$, and optimized-frame $F_{\text{avg}} = 0.731302$. The buffered channel gives $D_{\min} = 0.977156$, $u = 0.969593$, $V = 0.954637$, and optimized-frame $F_{\text{avg}} = 0.992331$. Its uncorrected identity fidelity is only 0.757738 because the off-diagonal PTM entries describe a large coherent rotation; calibration removes that deterministic frame error. Both Choi matrices are positive at numerical

precision ($\lambda_{\min} = 6.75 \times 10^{-2}$ direct and 2.37×10^{-5} buffered), and both trace-preservation residuals are below 9×10^{-16} .

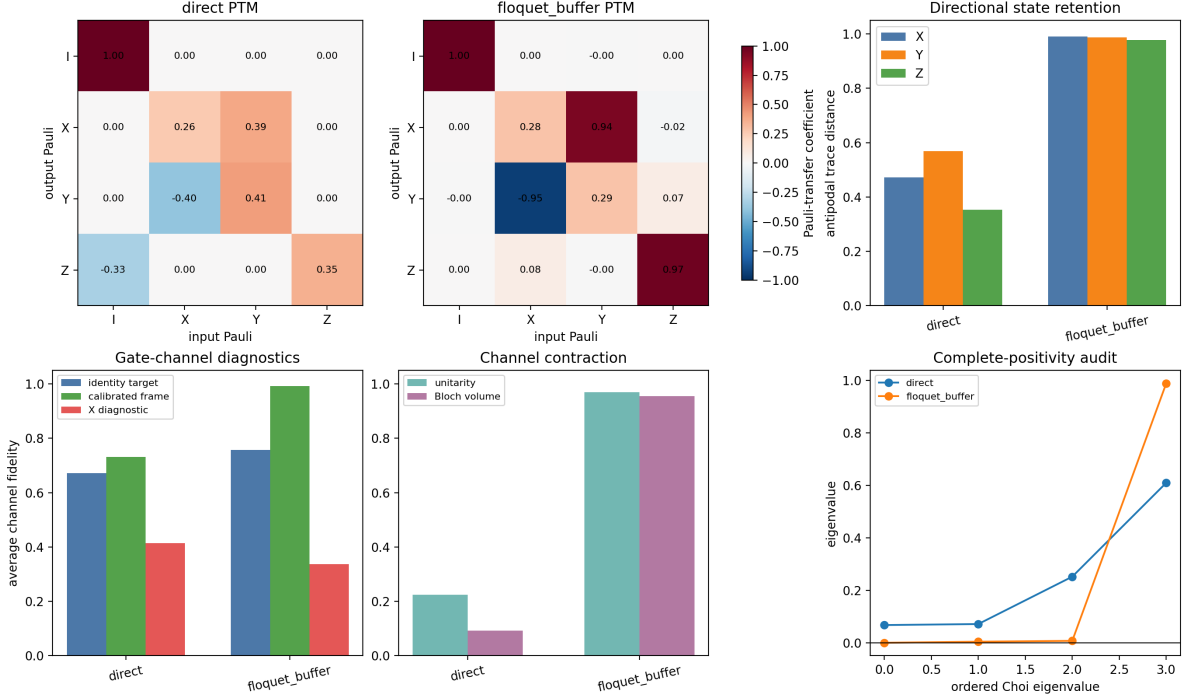


Figure 17: Six-state HEOM process tomography. The PTMs expose coherent rotation and irreversible contraction separately; the remaining panels compare directional retention, raw and calibrated average fidelity, unitarity, Bloch-volume contraction, and Choi positivity.

Information backflow and cancellation-safe power. For any initial pair, contractivity of a CP-divisible map requires $\dot{D}(t) \leq 0$. The Breuer–Laine–Piilo measure is

$$\mathcal{N}_{\text{BLP}} = \max_{\rho_1, \rho_2} \int_{\dot{D} > 0} \dot{D}[\rho_1(t), \rho_2(t)] dt. \quad (51)$$

We do not claim the continuous maximization in Eq. (51); summing positive increments for the three Pauli-antipodal pairs gives a reproducible lower bound. It is 0.021490 for direct contact and 3.38×10^{-4} for the buffer. Thus the buffer advantage here is not caused by stronger information backflow. It comes from spectral isolation: the integrated z -axis distinguishability increases from 3.07861 to 4.95044 while sampled backflow decreases. Figure 18 also plots $P(t)$ and the two non-cancelling cumulative work definitions from Eq. (46).

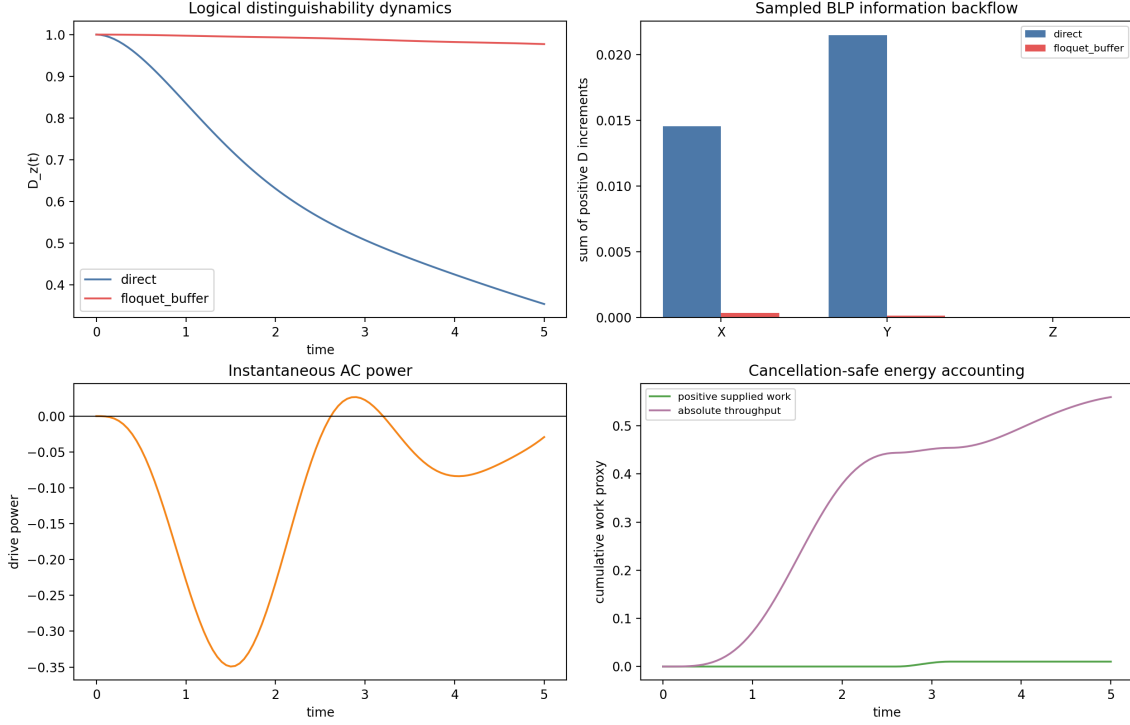


Figure 18: HEOM distinguishability, sampled BLP information backflow, and drive-power accounting. Absolute throughput remains finite even when positive and negative work nearly cancel in the net integral.

Readout phase and calibration uncertainty. The stroboscopic experiment was repeated for 16 uniformly spaced drive phases. The gain remains in $[0.618890, 0.623372]$ with median 0.620839, so the selected point does not require phase-perfect readout. A deterministic Latin-hypercube ensemble then perturbed amplitude by $\pm 10\%$, frequency by $\pm 5\%$, system-buffer coupling, reorganization energy, and bath cutoff by $\pm 15\%$, temperature by $\pm 10\%$, and phase by $\pm \pi/6$. All 48/48 HEOM samples satisfy $D_{\text{buffer}} \geq 0.75$ and $\Delta D > 0$. The median gain is 0.617616 and the empirical 5–95% interval is $[0.567750, 0.665942]$. A standardized local linear fit ranks reorganization energy, bath temperature, and bath cutoff as the dominant uncertainties; this fit is a sensitivity summary, not a global surrogate model.

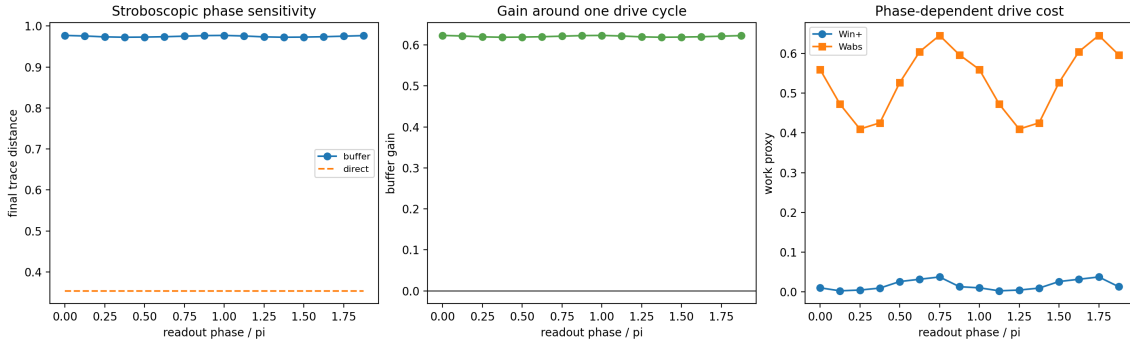


Figure 19: Stroboscopic drive-phase sensitivity of final distinguishability, buffer gain, and the two non-cancelling drive-work measures.

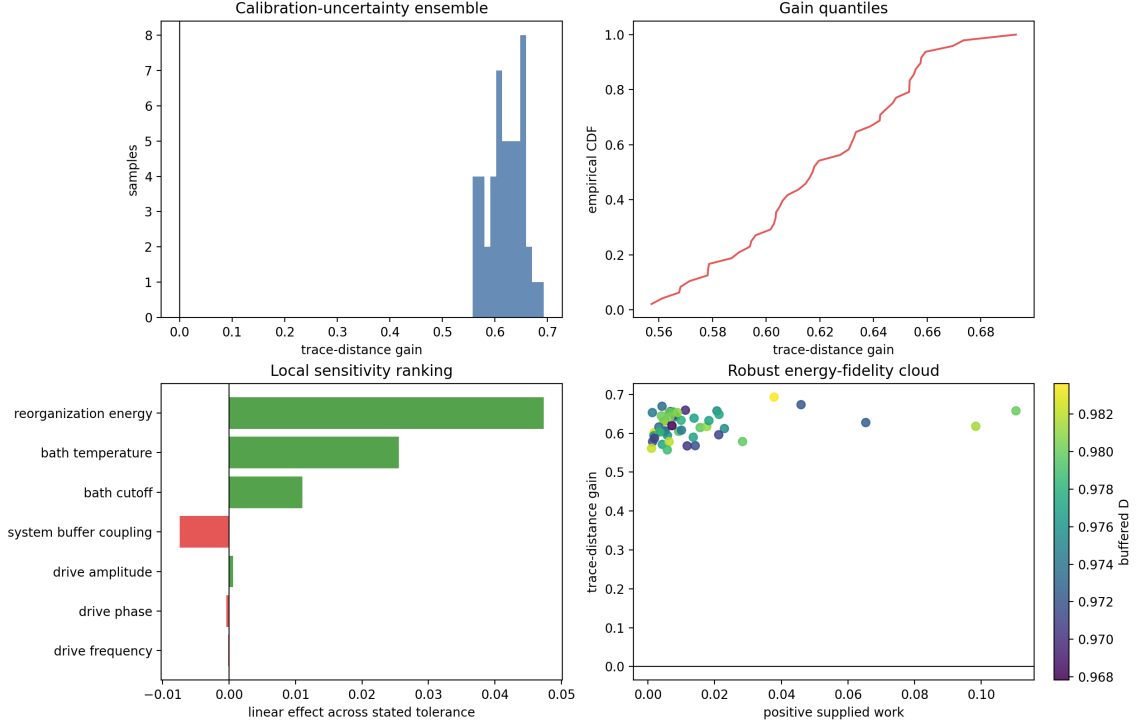


Figure 20: Forty-eight-point HEOM calibration-uncertainty ensemble. The histogram and empirical CDF quantify gain variation; the signed effects rank local sensitivities; the final panel shows the robust energy–fidelity cloud.

Two-axis numerical convergence. Hierarchy depth alone is insufficient because the bath correlation expansion also has to converge. We therefore scan $N_{\text{tier}} = 2, 3, 4$ and $N_k = 1, 2, 3$, defining

$$\delta_{N,K} = \max_{s \in \{0,1\}} D\left(\rho_s^{(N,K)}(t_f), \rho_s^{(4,3)}(t_f)\right). \quad (52)$$

At the production setting $(N, K) = (3, 2)$, $\delta_{3,2} = 3.289 \times 10^{-3}$ for direct contact and 1.322×10^{-4} for the buffered branch. The refined result therefore preserves the sign and scale of the 0.623 buffer advantage by a wide margin.

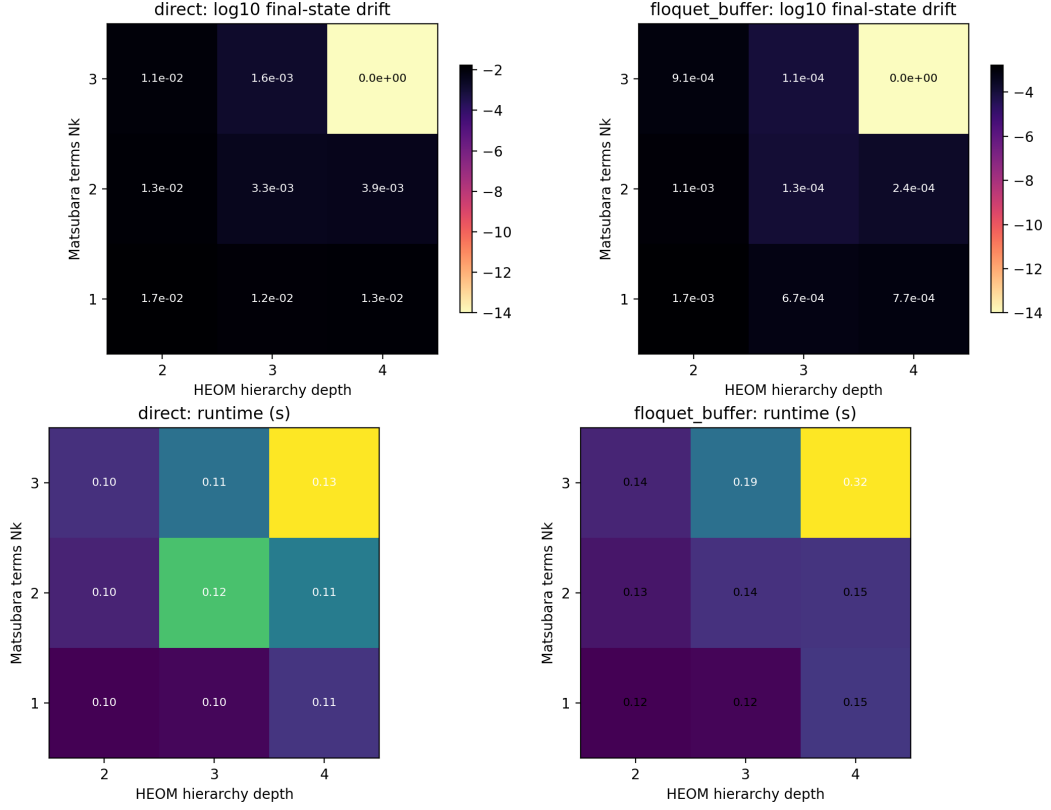


Figure 21: Joint convergence in HEOM hierarchy depth and Matsubara expansion order. The upper panels show final-state trace-distance drift relative to $(N_{\text{tier}}, N_k) = (4, 3)$; the lower panels report measured runtime.

Table 6: Approximations and scope limits specific to the advanced HEOM experiments.

Item	What was assumed	How the claim is limited or checked
Bath model	Drude–Lorentz spectral density represented by a finite Matsubara expansion.	Both hierarchy depth and Matsubara order are refined against $(4, 3)$.
Channel reconstruction	Linearity is used to reconstruct a one-qubit map from six Pauli eigenstates.	Choi positivity and trace preservation are checked numerically.
BLP witness	Only the three Pauli-antipodal pairs are sampled.	The reported value is a lower bound, not the globally optimized BLP measure.
Uncertainty ensemble	Independent bounded calibration perturbations and a first-order sensitivity fit.	The 48 points establish local robustness only inside the stated tolerance box.
Drive energy	$P(t) = \text{Tr}[\rho(t)\partial_t H(t)]$ is a quantum work proxy.	W_{net} , W_+ , and W_{abs} are all reported; cryogenic control-line losses are not included.
Gate scope	HEOM acts on the reduced output plus optional driven buffer.	It validates an output-stage channel, not a full six-qubit HEOM truth table.

Path integrals, Lamb shift, and QED implementation. The same Drude–Lorentz spectral-density model also gives the influence kernel used in the path-integral description and the dispersive principal-value shift that appears as a Lamb-shift correction in the GKSL/HEOM comparison. Fig. 22 shows the three supporting theory figures generated by code: the memory kernel $C(t) = \int J(\omega)[\coth(\beta\omega/2) \cos \omega t - i \sin \omega t]d\omega$, the structured-bath Lamb-shift proxy, and a circuit-QED style implementation sketch with an output qubit, driven buffer, lossy reaction-coordinate resonator, and engineered transmission-line bath.

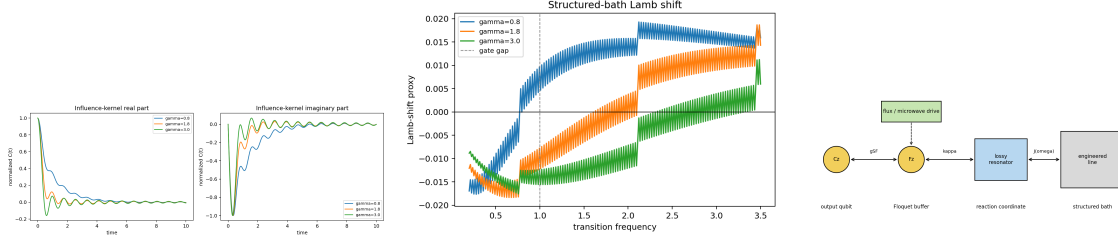


Figure 22: Supporting theory and implementation figures. Left: path-integral influence-memory kernel for different bath cutoffs. Middle: structured-bath Lamb-shift proxy showing drive/bath-induced detuning risk near the gate gap. Right: practical circuit-QED implementation route for the output buffer and engineered reservoir.

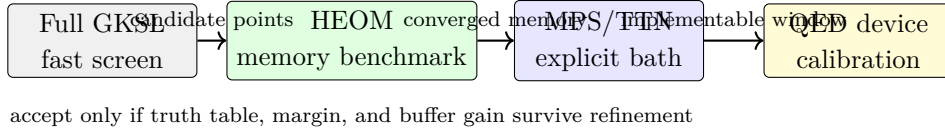


Figure 23: Validation pipeline used by the project: GKSL screens, HEOM checks structured-bath memory, MPS/TTN checks explicit-reservoir scaling, and the surviving window defines the practical QED implementation target.

10.3 Level 3: chain mapping with MPS or TTN

For the full structured-bath gate, the scalable route is to map each reservoir to a one-dimensional chain,

$$H_{B_\alpha} + H_{SB_\alpha} \longrightarrow \sum_{j=0}^{N_\alpha-1} \omega_{\alpha j} a_{\alpha j}^\dagger a_{\alpha j} + \sum_{j=0}^{N_\alpha-2} t_{\alpha j} (a_{\alpha j}^\dagger a_{\alpha, j+1} + \text{h.c.}) + V_\alpha (a_{\alpha 0} + a_{\alpha 0}^\dagger). \quad (53)$$

A one-branch direct benchmark is naturally an MPS if the bath chain is ordered after the gate degrees of freedom. For the full three-reservoir machine, both direct and buffered geometries are naturally factor-tree TTNs: the three-body interaction sits at the root, and the reference, input, and output bath chains form separate branches. The buffer only subdivides the output branch. This avoids forcing several independent bath correlations through one artificial MPS bond, as quantified in Section 11.

Backend	What it proves	Required convergence test
Full GKSL	Truth table and fast robustness map in the weak-memory limit.	Time step, Floquet period resolution, trace preservation, positivity, and agreement with reset equations when buffers are removed.
HEOM	Structured-bath memory benchmark for selected branches and selected full-gate points.	Hierarchy tier N_{tier} , expansion order K , Matsubara/Padé cutoff, and trace-norm convergence of ρ_0 .
MPS chain	Full many-body structured-bath evolution with explicit reservoirs.	Chain length N_α , bosonic cutoff d , bond dimension χ , TDVP/Trotter time step, and recurrence-reflection time.
TTN chain	Same physics as MPS, but with branch geometry matched to the gate.	Branch bond dimensions, root tensor dimension, local truncation errors, and comparison to MPS on small chains.

The gate is accepted only if all three statements hold: (i) the direct and buffered models reproduce the intended Boolean truth table; (ii) the buffered margin or distinguishability is positive after convergence; and (iii) the sign of the buffer advantage does not flip when moving from GKSL to HEOM and then to the MPS/TTN structured-bath simulation.

11 Tensor networks and graph geometry

The earlier single-branch calculation asks how large an MPS must be once an ordering has already been chosen. The full gate introduces a prior question: *which tensor-network graph should be used?* This is not cosmetic. The machine has three reservoir branches joined by one three-body interaction, so flattening it into a line can force otherwise independent branch correlations through the same virtual bond [13, 14]. This section derives that graph cost and measures it on exact evolved states.

11.1 Hamiltonian hypergraph, primal graph, and factor tree

Let the physical vertices be

$$V = \{C_0, C_1, C_z, F, B_{\alpha j} : \alpha \in \{0, 1, z\}\}, \quad (54)$$

where F is omitted for direct contact. Pair interactions give ordinary edges, but the gate term is the hyperedge

$$e_3 = \{C_0, C_1, C_z\}, \quad H_3 = g_3 \left(\sigma_+^{(0)} \sigma_-^{(1)} \sigma_-^{(z)} + \sigma_-^{(0)} \sigma_+^{(1)} \sigma_+^{(z)} \right). \quad (55)$$

There are two standard graph representations. The *primal graph* replaces e_3 by the triangle $(C_0, C_1), (C_0, C_z), (C_1, C_z)$. It contains one cycle and has treewidth 2. The *incidence* or factor graph instead introduces one nonphysical factor vertex U_3 and connects it separately to C_0 , C_1 , and C_z . After each reservoir is chain mapped, this incidence graph is a tree and therefore has treewidth 1.

This replacement is exact at the operator level rather than an approximation. Equation (55) is a sum of two product operators, so the central Hamiltonian factor has canonical polyadic operator rank 2. The passive buffer merely subdivides the output edge as $C_z - F - B_{z1}$. The Floquet term

$$H_D(t) = A \cos(\Omega t) \sigma_x^{(F)} \quad (56)$$

is local on F ; consequently it adds no spatial graph edge and cannot change treewidth. It can still change the numerical bond dimensions by creating more entanglement during time evolution.

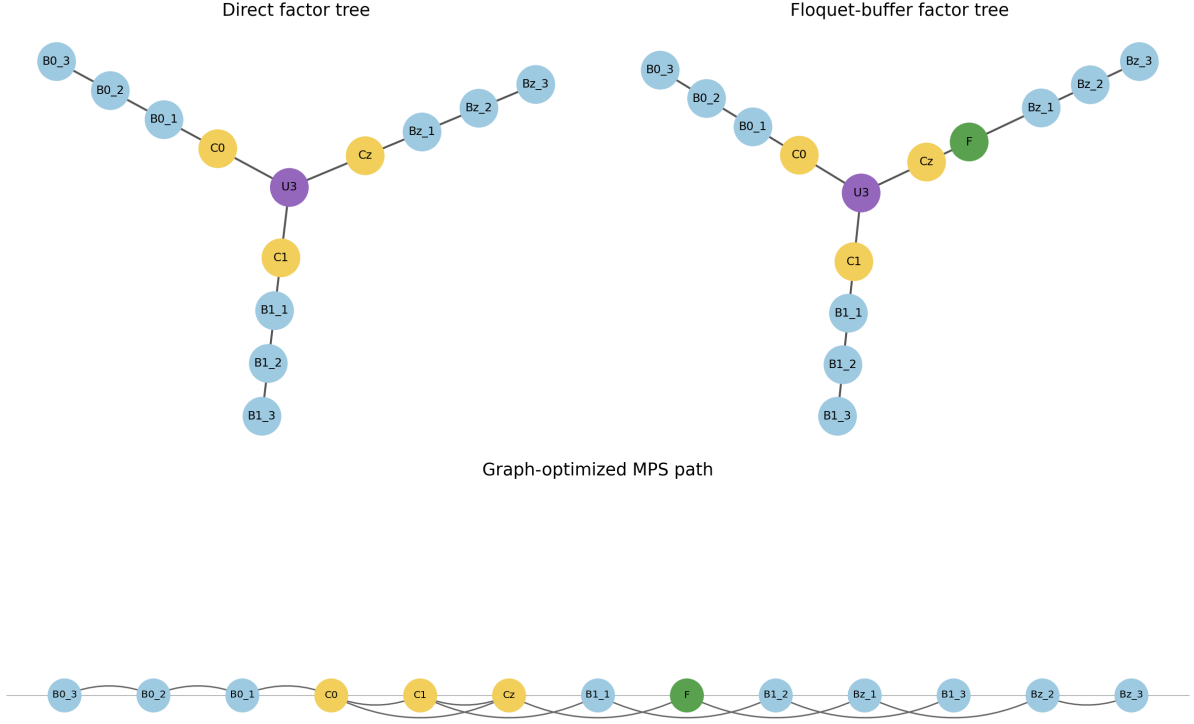


Figure 24: Tensor-network geometry of the three-reservoir machine. The purple U_3 tensor represents the three-body interaction. Direct and Floquet-buffered incidence graphs are trees; the buffer subdivides one branch without creating a cycle. The right panel shows a graph-optimized MPS ordering, where curved interaction edges reveal the price of forcing the branched graph onto one path.

11.2 Deriving the MPS ordering cost

An MPS first chooses a bijection $\pi : V \rightarrow \{1, \dots, n\}$. Its k th bond separates the prefix

$$S_k^\pi = \{v : \pi(v) \leq k\} \quad (57)$$

from its complement. For an interaction graph $G = (V, E)$, define the edge boundary and ordering cutwidth

$$\delta_G(S) = \{(u, v) \in E : u \in S, v \notin S\}, \quad (58)$$

$$c(\pi) = \max_{1 \leq k < n} |\delta_G(S_k^\pi)|. \quad (59)$$

The bandwidth

$$b(\pi) = \max_{(u,v) \in E} |\pi(u) - \pi(v)| \quad (60)$$

measures the longest interaction in the linear layout. Cutwidth controls how many interactions can create entanglement across one MPS bond, while bandwidth controls how many swap gates or long-range MPO channels are needed by a nearest-neighbour implementation. They are graph proxies, not substitutes for measuring the evolved state.

For a pure state, the Schmidt decomposition at cut k is

$$|\psi\rangle = \sum_{r=1}^{R_k} s_{kr} |L_{kr}\rangle |R_{kr}\rangle, \quad \sum_r s_{kr}^2 = 1. \quad (61)$$

Given an allowed discarded weight ε , the minimal numerical bond dimension is

$$\chi_k^*(\varepsilon) = \min \left\{ q : \sum_{r>q} s_{kr}^2 \leq \varepsilon \right\}, \quad \chi_{\text{MPS}}^* = \max_k \chi_k^*. \quad (62)$$

The bond entropy

$$S_k = - \sum_r s_{kr}^2 \log_2 s_{kr}^2 \quad (63)$$

obeys $S_k \leq \log_2 \chi_k$. For local evolution, entanglement can be generated across the cut only by terms in $\delta_G(S_k)$; schematically,

$$\left| \frac{dS_k}{dt} \right| \lesssim c_{\text{ent}} \sum_{e \in \delta_G(S_k)} \|h_e\| \log d_e. \quad (64)$$

Thus minimizing $c(\pi)$ is a principled ordering heuristic, although destructive interference and unequal couplings mean that the Schmidt spectrum remains the final numerical test.

If the MPS bond dimensions are $\chi_0 = \chi_n = 1$ and $\chi_1, \dots, \chi_{n-1}$, its raw tensor-entry count is

$$P_{\text{MPS}} = \sum_{k=1}^n d_k \chi_{k-1} \chi_k. \quad (65)$$

This count is representation dependent and includes gauge redundancy, but it is a useful memory and contraction-cost proxy when comparing two topologies for the same state.

11.3 Why a topology-matched TTN is different

For the factor-tree topology T , removing an edge e partitions the physical leaves into A_e and $\bar{A}_e$. The smallest exact TTN bond on that edge is the Schmidt rank of this bipartition; with truncation,

$$\chi_e^*(\varepsilon) = \min \left\{ q : \sum_{r>q} s_{er}^2 \leq \varepsilon \right\}. \quad (66)$$

Every reservoir branch is disconnected from the rest by one factor-tree edge, so its graph boundary is one independently of chain length. A linear MPS cannot place all three branches on opposite sides of the central interaction simultaneously. The TTN therefore trades the path bottleneck for a higher-degree root tensor. Its raw entry count is

$$P_{\text{TTN}} = \sum_{v \in T} d_v \prod_{e \ni v} \chi_e, \quad d_{U_3} = 1, \quad d_v = 2 \text{ for a spin site.} \quad (67)$$

The root product in Eq. (67) is the principal TTN cost. A TTN is advantageous only if the reduction of branch bond dimensions compensates for this product; the exact-state experiment below checks that condition directly.

11.4 Exact graph-state experiment and Quimb MPS compression

The reproducible code in `Scripts/tensor_network_graph_analysis/` performs two tests. First, it builds direct, passive-buffer, and Floquet-buffer graphs for $L = 1, \dots, 10$, evaluates an interleaved ordering, and searches deterministically over breadth-first, depth-first, reverse Cuthill–McKee, reversed, and local-swap orderings. At $L = 10$ all three architectures have primal-treewidth upper bound 2 and factor-treewidth 1. The interleaved MPS has cutwidth 4, the optimized MPS has cutwidth 3, and the factor-tree TTN has separator width 1. These widths remain constant with L because extending a branch adds vertices but no new junction.

Second, a state-level benchmark uses the finite spin Hamiltonian

$$H_{\text{geom}}(t) = \frac{1}{2} \sum_v \omega_v \sigma_z^{(v)} + H_3 + \sum_{(u,v) \in E_2} J_{uv} \sigma_x^{(u)} \sigma_x^{(v)} + \mathbf{1}_{\text{Floquet}} A \cos(\Omega t) \sigma_x^{(F)}. \quad (68)$$

The chosen dimensionless values are $(\omega_0, \omega_1, \omega_z, \omega_F) = (2, 1, 1, 1)$, $\omega_{B_j} = 0.82 + 0.04j$, $g_3 = 0.18$, contact coupling 0.14, C_z – F coupling 0.12, $J_j = 0.11/[1 + 0.15(j - 1)]$, $A = 0.45$, and $\Omega = 1.3$. Sparse exact propagation uses 24 midpoint steps over $0 \leq t \leq 4$. The largest calculations contain 15 physical spins for direct contact and 16 for buffered contact. Schmidt spectra are sampled every four steps using discarded-weight tolerance $\varepsilon = 10^{-10}$. The final state is also factorized and compressed as an MPS using Quimb [15].

Table 7: Maximum tolerance-controlled tensor requirements for exactly evolved states over the $L = 4$ time window. P is the maximum raw tensor-entry count. $F_{\chi=8}$ is the final-state fidelity after graph-ordered Quimb MPS compression.

Architecture	χ_{int}	χ_{opt}	χ_{TTN}	P_{opt}	P_{TTN}	$F_{\chi=8}$
Direct	61	23	10	4332	1702	0.9999693
Passive buffer	66	23	10	4650	1740	0.9999745
Floquet buffer	65	25	10	4986	1646	0.9999718

The ordering effect is already substantial: relative to interleaving, graph optimization reduces the maximum required MPS bond from 61 to 23 for direct contact and from 65 to 25 for the Floquet buffer. The topology-matched TTN reduces the maximum to 10. Its raw entry count is smaller than optimized MPS by 60.7% (direct), 62.6% (passive), and 67.0% (Floquet). The Floquet drive leaves all graph widths unchanged but raises the observed optimized-MPS maximum from 23 for the passive buffer to 25 for the driven buffer. Geometry and dynamics are therefore distinct sources of cost.

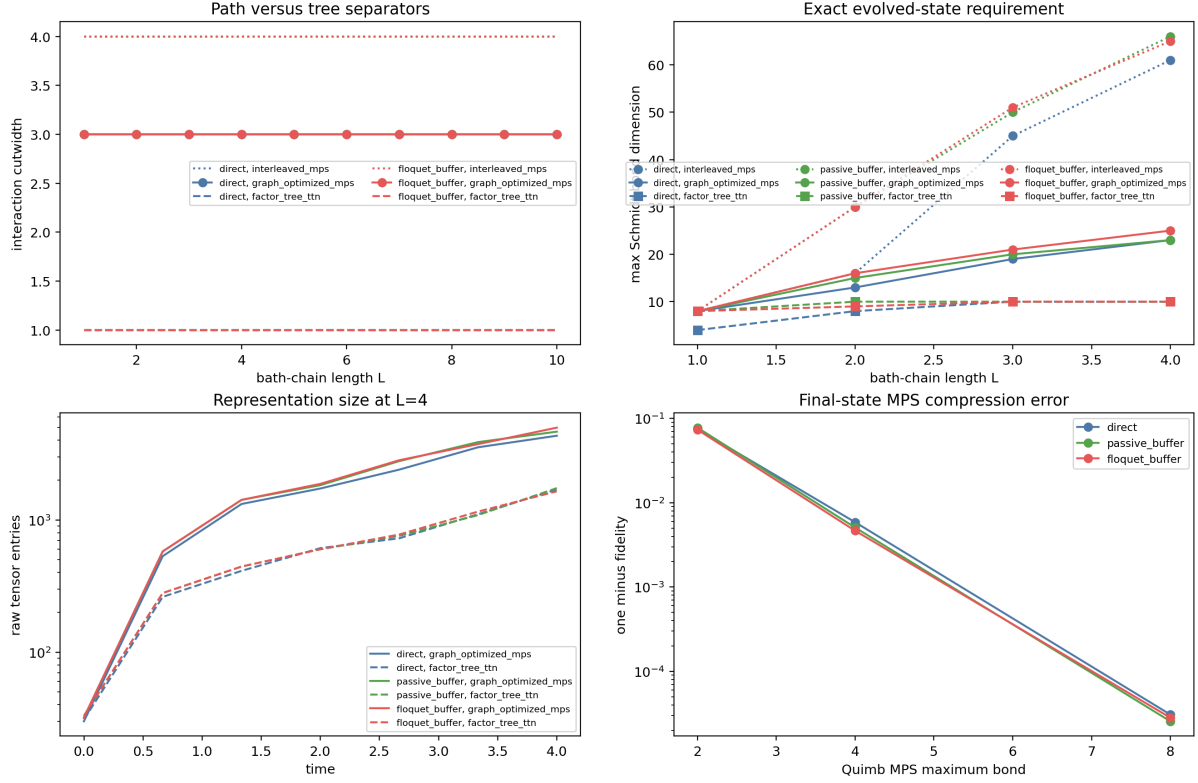


Figure 25: Graph and exact-state tensor-network benchmark. Top left: MPS cutwidth depends on ordering, while every factor-tree TTN edge is a one-edge separator. Top right: maximum required Schmidt bond over the exact time window. Bottom left: raw tensor entries at $L = 4$; dashed TTN curves remain below optimized MPS curves. Bottom right: graph-ordered Quimb MPS compression error, showing that $\chi = 8$ already retains more than 0.9999 fidelity despite the stricter 10^{-10} discarded-weight ranks reported in the table.

11.5 What this establishes and what remains

The calculation establishes four points. First, the natural mathematical object is a hyper-graph/factor graph, not a single bath chain. Second, site ordering is an algorithmic variable: a poor ordering can cost more than adding the physical buffer. Third, the factor-tree TTN is the lowest-rank exact representation tested for the full three-branch geometry. Fourth, an optimized MPS remains an excellent validation backend because a modest $\chi = 8$ gives final-state fidelity above 0.9999 in all three cases.

This is deliberately not presented as a converged thermodynamic-gate simulation. Approximations (A12)–(A14) replace bosonic modes by spins, use a pure product preparation, stop before long-chain steady-state physics, and factorize exact small-system states instead of evolving the tensor network variationally. The next tensor-network experiment should therefore use orthogonal-polynomial chain coefficients, thermofield purification of each thermal bath, two-site TDVP or TEBD, and independent convergence in $(L, d, \chi, \Delta t)$. The graph result gives a concrete backend decision for that calculation: use MPS for single branches and cross-checks, but use the factor-tree TTN as the primary full-gate geometry.

12 Architecture decision and CMOS–inverter correspondence

12.1 Three realizable bath-contact architectures

The numerical word “buffered” contains two different physical ideas: inserting an intermediate subsystem and driving that subsystem. They must be separated before asking whether the Floquet architecture is actually better. For one output operator S and bath operator B , the three Hamiltonian topologies are

$$H_{\text{direct}} = H_M + H_B + g_{MB} S \otimes B, \quad (69)$$

$$H_{\text{passive}} = H_M + H_F + g_{MF} S \otimes F + H_B + g_{FB} F \otimes B, \quad (70)$$

$$H_{\text{Floquet}}(t) = H_{\text{passive}} + A \cos(\Omega t + \phi) F_x. \quad (71)$$

Here M denotes the integrated autonomous thermal machine. The direct and passive Hamiltonians are time independent and therefore strictly autonomous once the thermal reservoirs are connected. The passive subsystem F may be a reaction-coordinate resonator, auxiliary qubit, or impedance filter. The Floquet device is a hybrid thermal machine: its heat currents still perform the logic, but an external phase reference and work source continuously modulate F .

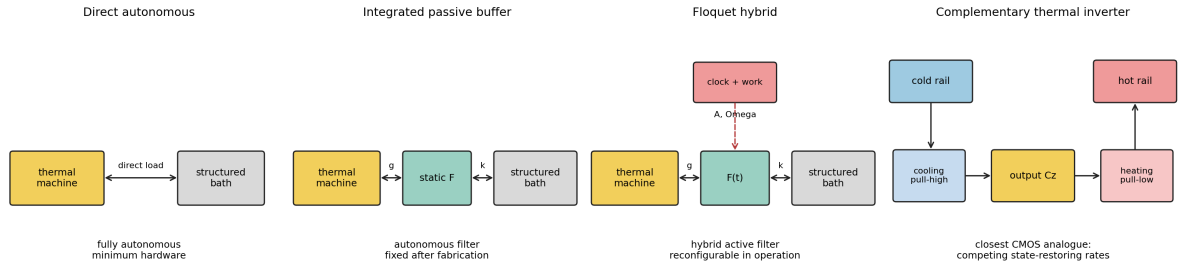


Figure 26: Four architecture classes. Direct contact minimizes hardware; the static integrated buffer preserves autonomy while filtering the bath; the Floquet hybrid adds in-operation spectral control; and a genuinely CMOS-like thermal inverter requires complementary pull-high and pull-low dissipative channels rather than one buffer alone.

For matched bath and coupling parameters, define the exact decomposition

$$\underbrace{D_F(A) - D_{\text{direct}}}_{\text{total buffered gain}} = \underbrace{D_F(0) - D_{\text{direct}}}_{\text{passive isolation}} + \underbrace{D_F(A) - D_F(0)}_{\text{drive-specific Floquet gain}}. \quad (72)$$

At the best-gain HEOM bath point $(g_{SF}, \lambda, \gamma_c) = (0.04, 0.075, 0.8)$, direct contact gives $D_{\text{direct}} = 0.353784$, the static integrated buffer gives $D_F(0) = 0.966563$, and the best driven setting gives $D_F(0.55) = 0.977156$. Therefore

$$\Delta D_{\text{total}} = 0.623372 = 0.612779 + 0.010593. \quad (73)$$

Approximately 98.3% of the total improvement at this point comes from inserting the intermediate subsystem; only 1.7% is the incremental effect of driving it. The driven increment costs $W_{\text{abs}} = 0.559420$ in the dimensionless work-throughput convention of Eq. (46).

This single point does not make the drive irrelevant. Repeating the matched comparison for all 243 driven HEOM points gives

$$D_F(A) - D_F(0) > 0 \quad \text{for } 243/243 \text{ points}, \quad (74)$$

with median increment 0.020150, empirical 5–95% interval [0.002450, 0.164534], and maximum increment 0.263747. Larger drive amplitudes have larger median increments, and the largest gains occur when the static buffer is less well matched. Thus periodic control is most defensible as a *tunable rescue and calibration layer*, not as the sole origin of isolation.

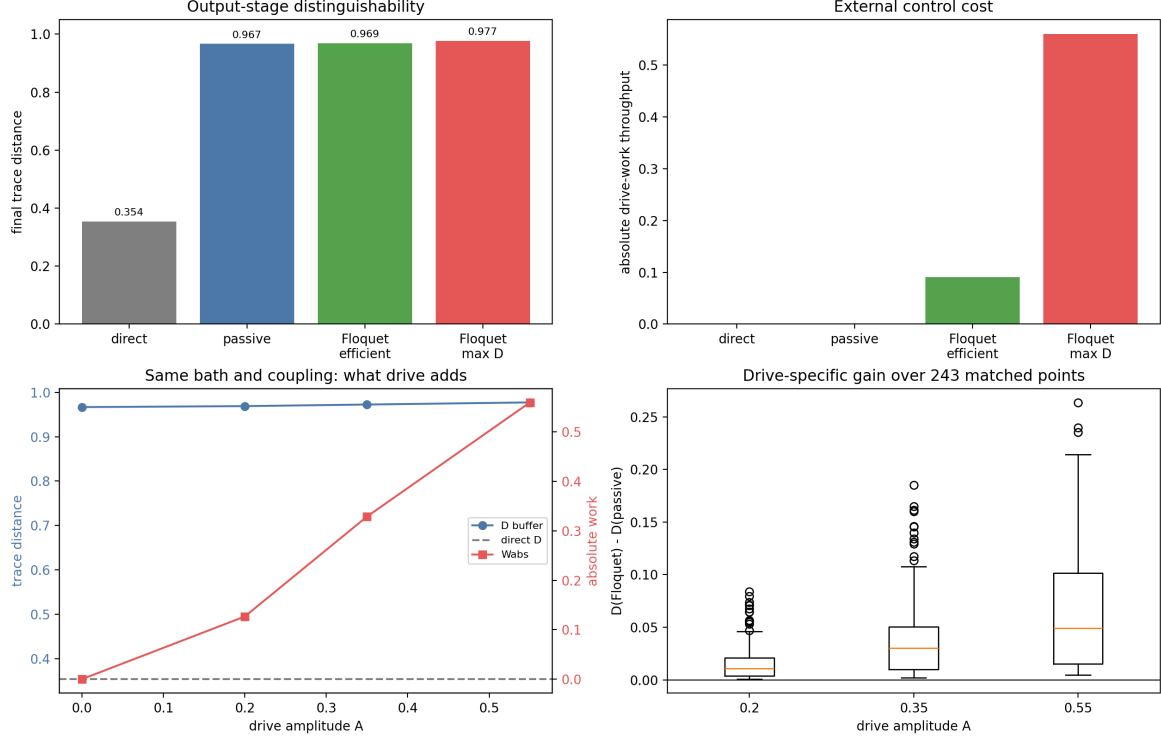


Figure 27: Architecture-level HEOM comparison. Top: distinguishability and external drive throughput for direct, passive, and two driven operating points. Bottom left: adding drive at one fixed bath/coupling point gives a small increase in D while work grows. Bottom right: distribution of the drive-specific increment over all 243 matched driven/passive pairs.

The earlier reduced bridge ablation found that its default undriven spacer degraded distinguishability, whereas the broader HEOM sweep finds strong passive-buffer performance at weak g_{SF} . These results are not mathematically contradictory: the Hamiltonians, bath descriptions, and parameter points differ. Their combination establishes the more useful claim that a static buffer is *not automatically* beneficial; it must be impedance- and frequency-matched, while Floquet control expands the set of parameters that can be repaired after fabrication.

12.2 Is the Floquet architecture actually better?

There is no architecture-independent yes/no answer because “better” requires an objective function. A minimal decision functional is

$$\mathcal{C}_a = w_D(1 - D_a) + w_W W_{\text{abs},a} + w_C C_{\text{control},a} + w_H C_{\text{hardware},a}, \quad (75)$$

where the weights encode whether a device values logical fidelity, external work, control electronics, or component count. The computed results support the following conditional decision.

Table 8: Architecture decision from the present simulations. Numerical D values refer to the same selected reduced output-stage HEOM point and are not a full-device energy comparison.

Criterion	Direct contact	autonomous	Integrated buffer	passive	Floquet buffer
Final D at selected bath	0.353784		0.966563		0.977156
External drive throughput	0		0		0.559420 at maximum- D point
Strict autonomy	Yes		Yes		No: clock and work source required
Spectral isolation	None beyond bare coupling		Fixed H_F, g_{MF}, g_{FB}	by	Sideband-tunable during operation
Fabrication/calibration	Lowest count	component	Extra resonator/qubit and fixed matching		Extra subsystem, drive line, phase and Lamb-shift calibration
Best use case	Benign bath; minimum hardware	Markovian	Stable structured bath known at design time		Unknown/drifting spectrum or post-fabrication retuning
Current evidence	Lowest D in point	work but weak hard-memory	Captures most of the HEOM gain		Improves every matched passive point, often modestly

Hence the *integrated passive buffer is the preferred default architecture under the present fixed-parameter evidence*: it preserves autonomy and obtains nearly all of the best-point distinguishability. The Floquet buffer is better when its reconfigurability has operational value—for example compensating fabrication disorder, tracking a moving bath resonance, widening a robust operating window, or shortening a calibration cycle. Direct contact remains rational when the reservoir is already weak and Markovian, component count dominates, or any external clock is disallowed. The Floquet branch therefore does not dominate the autonomous alternatives in the Pareto sense: it increases D but also increases work and control complexity.

This conclusion is intentionally limited to the reduced HEOM output stage. A final device-level ranking needs the full gate truth table, steady heat currents, entropy production, reset cost, drive-line loss, and fabrication overhead evaluated in one common model. In particular, W_{abs} is not the wall-plug power of a cryogenic microwave source.

12.3 Deriving the CMOS comparison

The analogy to a CMOS inverter is useful only after specifying the transfer law. In a symmetric long-channel square-law model, define

$$I(\Delta, V_D) = k \begin{cases} 0, & \Delta \leq 0, \\ \Delta V_D - \frac{1}{2} V_D^2, & 0 < V_D < \Delta, \\ \frac{1}{2} \Delta^2, & V_D \geq \Delta, \end{cases} \quad (76)$$

with a small channel-length-modulation correction $I \mapsto I(1 + \lambda_{\text{ch}} V_D)$. For the nMOS and pMOS branches,

$$I_n = I(V_{\text{in}} - V_T, V_{\text{out}}), \quad I_p = I(V_{DD} - V_{\text{in}} - V_T, V_{DD} - V_{\text{out}}), \quad (77)$$

and the DC transfer curve is obtained from $I_n = I_p$. The complementary devices actively connect the output toward one of two stiff voltage rails; near threshold both conduct, producing large negative gain $-dV_{\text{out}}/dV_{\text{in}}$.

A complementary thermal inverter requires the analogous competition of two input-controlled dissipative rates, not merely a subsystem placed between output and bath. If p is the excited-state population,

$$\dot{p} = \Gamma_{\uparrow}(\beta_{\text{in}})(1 - p) - \Gamma_{\downarrow}(\beta_{\text{in}})p, \quad p_{\infty} = \frac{\Gamma_{\uparrow}}{\Gamma_{\uparrow} + \Gamma_{\downarrow}}, \quad (78)$$

and

$$\beta_{\text{out}} = \frac{1}{\epsilon_z} \ln\left(\frac{1 - p_{\infty}}{p_{\infty}}\right). \quad (79)$$

The cooling channel connected to the cold, high- β rail is the logical pull-high branch; the heating channel connected to the hot, low- β rail is the pull-low branch. The input must tune these rates in opposite directions. Floquet sidebands can tune $\Gamma_{\uparrow, \downarrow}$, but the periodic drive alone does not create complementary restoration unless both channels are engineered.

For normalized input and output

$$x = \frac{\beta_{\text{in}} - \beta_{\text{hot}}}{\beta_{\text{cold}} - \beta_{\text{hot}}}, \quad y = \frac{\beta_{\text{out}} - \beta_{\text{hot}}}{\beta_{\text{cold}} - \beta_{\text{hot}}}, \quad (80)$$

the small-signal restoration gain is $G = |dy/dx|$. Define x_{IL} and x_{IH} by the two crossings $G = 1$, and let $y_H = y(x_{IL})$, $y_L = y(x_{IH})$. The normalized static noise margins are

$$NM_L = x_{IL} - y_L, \quad NM_H = y_H - x_{IH}. \quad (81)$$

The idealized CMOS calculation gives $(NM_L, NM_H) = (0.356, 0.356)$; the reconstructed thermodynamic NOT at $\epsilon_1 = 20$ gives $(0.310, 0.295)$ and maximum normalized slope $G_{\text{max}} = 4.995$. The CMOS peak slope (~ 115 in the discretized ideal square-law model) is model-sensitive and is shown only to indicate a much sharper switching region, not as a measured technology comparison.

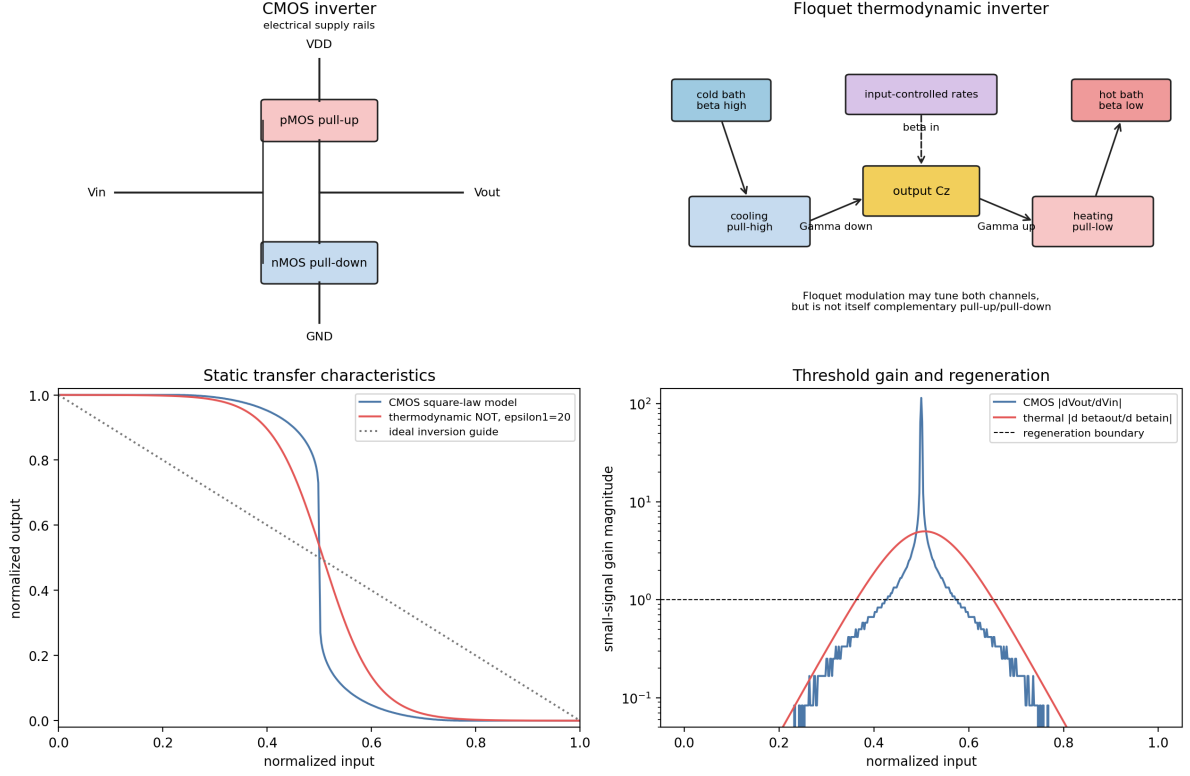


Figure 28: CMOS and thermodynamic-inverter comparison generated from code. Top: complementary electrical pull-up/pull-down and the corresponding cold-rail cooling / hot-rail heating architecture. Bottom: normalized transfer curves and small-signal gain. The thermodynamic NOT has a regenerative $G > 1$ region, but the present Floquet buffer is a tunable spectral stage rather than a complete complementary inverter.

Table 9: CMOS dictionary and the limits of the analogy.

CMOS quantity	Thermodynamic counterpart	Important limitation
V_{in}	Input bath inverse temperature or virtual temperature	Temperature is not an instantaneous node voltage.
V_{out}	Output β_{out} , polarization, or decoded population	Usually inferred statistically from repeated measurements.
V_{DD}/ground	Cold high- β / hot low- β logical rails	Rails are reservoirs with heat and entropy currents.
pMOS/nMOS	Cooling pull-high / heating pull-low dissipative channels	Hamiltonian couplings are generally bidirectional, not ideal unilateral switches.
Voltage gain	$ d\beta_{out}/d\beta_{in} $ or normalized G	Trace-distance preservation is a separate quantum-channel metric.
Noise margin	Distance from the thermal decoding threshold	Depends on bath fluctuations, finite heat capacity, and readout statistics.
Fan-out	One decoded thermal output biasing later gates	Arbitrary quantum states cannot be copied; practical cascading is classical or dissipative.
Supply power	Steady heat currents plus any Floquet work	Must include entropy production, reset, and control-line losses.

The correct engineering statement is therefore narrower than “Floquet buffer equals CMOS inverter.” Both can isolate a logical node, shape a sigmoidal inversion curve, and improve noise margin through a regenerative region. CMOS already integrates complementary restoration, directional gain, stiff rails, and mature fan-out. The current Floquet buffer supplies spectral isolation and tunability; it becomes genuinely CMOS-like only when embedded in a two-channel thermal architecture whose opposing rates restore both logical rails and whose output remains stable under a specified downstream load.

13 Robustness programme and outlook

The remaining scientific task is to map the region where the buffered thermodynamic NOT stays correct *and* well separated. The parameter vector is $\theta = (\alpha, \omega_c, s, \tau_{\text{mem}}, \Delta t, A, \Omega, g_{SF}, \epsilon_F, t_f)$, and the robust region enforces a margin,

$$\mathcal{R}_{\text{robust}}(m_0) = \{\theta : \mathcal{A}_{\text{NOT}}(\theta) = 1, m(\theta) > m_0\}, \quad (82)$$

the central comparison being $\mathcal{R}_{\text{robust}}^{\text{buffered}}(m_0)$ versus $\mathcal{R}_{\text{robust}}^{\text{direct}}(m_0)$. Each point is classified as fail ($\mathcal{A} < 1$), fragile pass ($\mathcal{A} = 1, m < m_0$), or robust pass ($\mathcal{A} = 1, m \geq m_0$). A first GKSL surrogate sweep (80 points, $m_0 = 0.05$) gave 68 robust passes, 0 fragile, 12 fail (Figs. 29–31); however the best buffered–minus–direct margin gain was only 4.02×10^{-5} , so in this surrogate the buffer preserves the truth table without yet enlarging the margin. The reduced output-stage HEOM sweep now establishes a positive drive-specific distinguishability increment at all 243 matched points, but it does not propagate the full three-qubit Boolean gate. Therefore enlargement of the *full-gate* robust region still requires the PT-MPO/TEMPO or a full-machine HEOM backend.

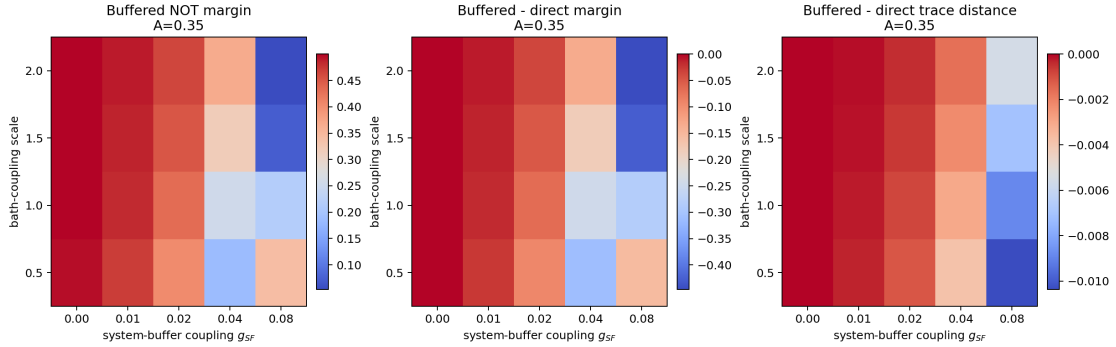


Figure 29: GKSL robustness phase maps at the largest drive amplitude: buffered margin, buffered–direct margin, and trace–distance gain over system–buffer and bath–coupling scales.

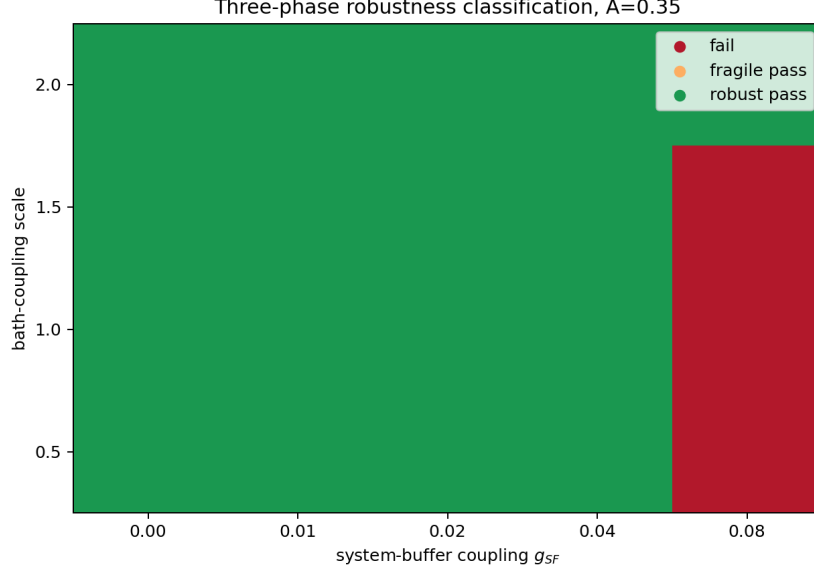


Figure 30: Three-phase classification (robust pass / fragile / fail), analogous to CMOS noise-margin analysis.

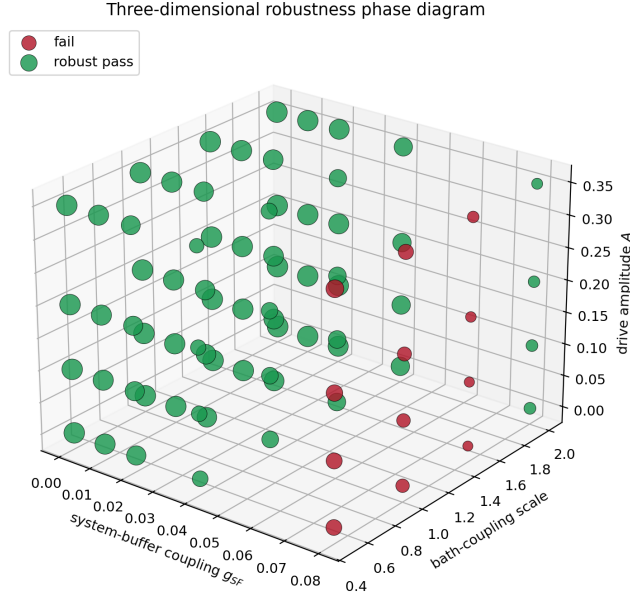


Figure 31: Three-dimensional robustness phase diagram over drive amplitude, bath-coupling scale, and system-buffer coupling; marker size scales with the buffered decoding margin.

Validation required for a journal-grade claim. The present output-stage HEOM work supplies hierarchy convergence, an independent structured-bath benchmark, disorder sampling, energy bookkeeping, and a trace-distance-backflow diagnostic. Before asserting that the buffer enlarges the *full-gate* robust region under genuine memory, the programme must still add: (i) a weak-coupling cross-check that PT-MPO reproduces the GKSL limit; (ii) a full-gate PT-MPO convergence surface $(\Delta t, \tau_{\text{mem}}, \epsilon_{\text{SVD}}) \mapsto (\mathcal{A}_{\text{NOT}}, m, \max_t \delta_{\text{tr}}, \max_t \delta_{\text{herm}})$; (iii) converged MPS/TTN propagation of the complete gate and reservoirs; (iv) threshold cross-validation on held-out bath parameters; and (v) a full-device strong-coupling audit including heat, interaction

energy, reset, and drive-line losses. A robust claim requires the same truth table and the same sign of the buffer advantage to survive refinement of these controls.

Outlook. The staged evidence is: an analytic baseline that reconstructs the NOT mechanism; a three-qubit GKSL surrogate that passes the direct and buffered truth tables; and a corrected PT-MPO/TEMPO structured-bath backend that also passes a calibrated truth table but with a small buffered margin; and a converged reduced HEOM output model whose buffer and drive contributions are now separated quantitatively. The project has thus entered the full-gate robustness-mapping stage. Output-stage drive work has been audited in the stated convention, while full-device heat, entropy, reset, and wall-plug claims remain provisional. The rigorous driven-dissipative foundation for that programme — the periodic GKSL generator, the chain/MPS treatment, and the work-distinguishability conjecture — is the subject of Part II.

Table 1: Master list of approximations used in Part I, with the replacement each makes, its regime of validity, and the failure mode when it is violated. Tags are referenced throughout the text.

Tag	Approximation	Statement / replacement	Regime of validity & failure mode
(A1)	Factorized initial state	$\rho_{\text{tot}}(0) = \rho_S(0) \otimes \rho_B$, so $\mathcal{I}(t) = 0$ in (7)	Valid if system and bath start uncorrelated; else an inhomogeneous memory term appears.
(A2)	Born / weak coupling	$\rho_{\text{tot}}(t) \approx \rho_S(t) \otimes \rho_B$; H_{SB} bilinear	$\lambda \ll \epsilon$; fails when bath back-action and interaction energy are non-negligible.
(A3)	Markov / short memory	$\rho_S(\tau) \rightarrow \rho_S(t)$, $\int_0^t \rightarrow \int_0^\infty$	$\tau_B \ll \tau_R$; fails for structured reservoirs that return information (backflow).
(A4)	Secular / RWA	drop terms with $\omega \neq \omega'$	$\gamma \ll$ Bohr gaps; fails for near-degenerate transitions and drive sidebands.
(A5)	Reset thermalization	$\mathcal{L}^{(k)}\rho = \gamma_k(\text{Tr}_k \rho \otimes \tau(\beta_k) - \rho)$	Steady-state populations only; changes transient shape, not the inversion mechanism.
(A6)	Energy-preserving H_{int}	$[H_{\text{int}}, H_0] = 0$	$\chi \ll \epsilon$; off-resonant (energy-changing) terms otherwise leak.
(A7)	Local master equation	bare-qubit dissipators in (14)	$\chi \ll \epsilon$; else use the global (dressed) dissipator to avoid second-law violations.
(A8)	Modulator pinned to reference	$p_{\mathcal{M}}^{\text{ss}} \approx g_z(\beta_r)$	$\mu' \ll \gamma$; else the modulator is dragged by the output bath.
(A9)	Adiabatic elimination	steady currents inserted into the slow β_z ODE	$\mu, \mu' \ll \gamma$; error $O(\mu/\gamma)$; fails if timescales overlap.
(A10)	Small- ϵ_z sigmoid	$g_z(\beta) \approx \frac{1}{2} - \frac{\beta\epsilon_z}{4}$	$\epsilon_z \ll \epsilon_1$; larger ϵ_z blurs the parameter roles.
(A11)	Weak heat-current split	$\dot{Q}_\alpha = \text{Tr}(H_S \mathcal{D}_\alpha \rho)$	Weak coupling; reduced-system heat currents ambiguous at strong coupling.
(A12)	Spin-chain bath proxy	Each chain site has $d = 2$ instead of a converged bosonic cutoff	Tests tensor geometry and entanglement compression, not a quantitative bosonic reservoir.
(A13)	Finite graph/time window	$L \leq 4$ and $0 \leq t \leq 4$ in the exact graph benchmark	Valid before boundary reflections dominate; does not establish a periodic or thermodynamic steady state.
(A14)	Schmidt truncation	Retain the smallest χ with discarded weight at most 10^{-10}	Controlled state-compression criterion; observables still require a separate convergence test in χ .

Part II

Abhishek: Periodically Driven Open-System Logic Gates

14 Introduction

14.1 What is thermodynamic computing?

Thermodynamic computing [7] proposes that the natural relaxation of a physical system coupled to heat baths can be harnessed to perform computation. In the quantum regime, autonomous thermal machines composed of a few interacting qubits can act as “thermodynamic neurons”: logical inputs are encoded in the temperatures of finite-size heat baths, and the logical output is read from the temperature of an auxiliary output bath once the system has settled into a nonequilibrium steady state (NESS) [1, 8]. The key appeal is that such devices operate without an external clock: computation is driven entirely by the flow of heat from hot to cold reservoirs.

14.2 Why add periodic driving?

All existing quantum thermodynamic neurons are *autonomous*, i.e., their Hamiltonians are time-independent. In this note we ask a natural question: can one improve the speed or accuracy of a thermodynamic neuron by adding a periodic driving field? This is a form of *Floquet engineering*: a periodically driven auxiliary system (the *Floquet buffer*) is inserted between the logical qubits and the physical baths. The buffer acts as a tunable effective environment whose spectral density can be shaped by the drive.

The price is twofold. First, the device is no longer autonomous; it requires a classical clock to maintain the drive and to read out the result stroboscopically. Second, an external work input is needed to power the drive. Nevertheless, the resulting architecture—a *periodically driven dissipative logic gate*—is an interesting candidate for low-energy computation.

14.3 Scope and structure of this note

This note is written as a self-contained pedagogical resource for a collaborative project involving a theoretical physicist, a mathematician working on tensor networks, and an MSc student with a background in engineering and coding. We assume familiarity with basic quantum mechanics (Hilbert spaces, operators, partial trace, von Neumann equation) and statistical mechanics (canonical ensemble, partition function). All more advanced concepts (GKSL equation, Floquet theory, chain mapping of baths, matrix-product states) are reviewed in the main text or in the appendices.

The note is organised as follows.

- Section 15 provides concise reviews of the GKSL master equation, Floquet theory, the chain representation of bosonic baths, matrix-product states (MPS), and the trace distance as a distinguishability measure.
- Section 16 describes the Floquet-buffer architecture in detail.

- Section 17 states and proves a theorem (Proposition 17.1) that in a certain weak-coupling limit the logical qubits obey a time-periodic GKSL equation.
- Section 18 explains how to go beyond weak coupling using chain mapping and MPS, and how to extract energy currents and work without exponentially large bond dimensions.
- Section 19 proposes a conjectured work–distinguishability tradeoff.
- Section 20 outlines a staged numerical programme.
- The appendices contain additional technical details.

15 Background material

Here we collect, in a deliberately compact form, the mathematical tools that are essential for the rest of the note. The reader may skip this section on first reading and return as needed.

15.1 GKSL (Lindblad) master equation

A Markovian open quantum system is described by a density matrix ρ evolving under the GKSL (Gorini-Kossakowski-Sudarshan-Lindblad) equation:

$$\dot{\rho} = -i[H, \rho] + \sum_k \gamma_k \left(L_k \rho L_k^\dagger - \frac{1}{2} \{L_k^\dagger L_k, \rho\} \right), \quad (83)$$

where H is the effective system Hamiltonian, L_k are jump operators, and $\gamma_k \geq 0$ are positive rates. The generator is trace-preserving and completely positive. If the rates satisfy local detailed balance with respect to a heat bath at inverse temperature β , the system relaxes to a Gibbs state $\rho_\beta \propto e^{-\beta H}$ (when there is only one bath) or to a NESS when several baths are present.

15.2 Floquet theory and Floquet-Lindblad equation

A periodically driven system has a Hamiltonian $H(t) = H(t + T)$ with $T = 2\pi/\Omega$. The time-evolution operator can be written in terms of Floquet modes $|\phi_n(t)\rangle$ and quasi-energies ϵ_n :

$$U(t) = \sum_n e^{-i\epsilon_n t} |\phi_n(t)\rangle \langle \phi_n(0)|, \quad (84)$$

where $|\phi_n(t + T)\rangle = |\phi_n(t)\rangle$ and $\epsilon_n \in [-\Omega/2, \Omega/2)$.

When the system is open and weakly coupled to one or more baths, a Floquet-Lindblad equation can be derived [9, 10]. The generator becomes time-periodic:

$$\dot{\rho}(t) = \mathcal{L}(t)\rho(t), \quad \mathcal{L}(t + T) = \mathcal{L}(t). \quad (85)$$

The jump operators are expressed in the Floquet basis, and the rates depend on the Fourier components of the bath correlation functions at frequencies shifted by multiples of Ω . A crucial feature is the appearance of sidebands: transitions can occur at energies $\epsilon_m - \epsilon_n + q\Omega$, $q \in \mathbb{Z}$, which allows the system to absorb or emit quanta of the drive while exchanging energy with the baths.

15.3 Chain mapping of bosonic baths

A heat bath is usually modelled as a collection of independent harmonic oscillators:

$$H_B = \int_0^\infty d\omega \omega a_\omega^\dagger a_\omega, \quad (86)$$

coupled to the system by $H_{SB} = A \otimes \int d\omega \sqrt{J(\omega)}(a_\omega + a_\omega^\dagger)$, where $J(\omega)$ is the spectral density. For numerical purposes it is convenient to map this continuum model to a semi-infinite one-dimensional chain with nearest-neighbour hopping [11, 12]. The transformation uses a sequence of orthogonal polynomials (e.g., Lanczos or Chebyshev) determined by $J(\omega)$. After the mapping, the Hamiltonian becomes

$$H_{\text{chain}} = \sum_{k=1}^{\infty} \omega_k a_k^\dagger a_k + \sum_{k=1}^{\infty} t_k (a_k^\dagger a_{k+1} + a_{k+1}^\dagger a_k), \quad (87)$$

where a_k ($k \geq 1$) are new bosonic operators, and the system couples only to the first site: $H_{SB} \propto A \otimes (a_1 + a_1^\dagger)$. The parameters ω_k and t_k are determined recursively from the spectral density. For an Ohmic bath $J(\omega) \propto \omega$ with a suitable cutoff, the coefficients t_k decay rapidly with k , so that a truncation to a finite chain of length N (typically $N \sim 100 - 200$) gives converged results for the reduced system dynamics.

15.4 Matrix-product states (MPS) and the TDVP

A one-dimensional quantum lattice system (such as the chain-mapped bath) can be efficiently represented by a matrix-product state:

$$|\psi\rangle = \sum_{\{s_i\}} C^{s_1} A^{s_2} \cdots A^{s_N} |s_1 s_2 \dots s_N\rangle, \quad (88)$$

where C and A^{s_i} are matrices, and the bond dimension χ controls the amount of entanglement that can be captured. The time-dependent variational principle (TDVP) or Trotter-Suzuki decompositions allow one to evolve the MPS in real time under a Hamiltonian $H(t)$. The computational cost scales as $O(N\chi^3 d^2)$ per time step, where d is the local Hilbert space dimension.

For the chain-mapped bath at moderate temperature, the entanglement entropy between the system and the first few chain sites saturates at a finite value. Consequently, the required bond dimension χ does not grow with the chain length N ; values of $\chi \approx 30-50$ are typical for converged simulations.

15.5 Trace distance as a distinguishability measure

The trace distance between two density matrices ρ and σ is

$$D(\rho, \sigma) = \frac{1}{2} \|\rho - \sigma\|_1, \quad (89)$$

where $\|A\|_1 = \text{Tr} \sqrt{A^\dagger A}$. It satisfies $0 \leq D(\rho, \sigma) \leq 1$ and is a metric on the space of states. Operationally, $D(\rho, \sigma)$ is equal to the maximum success probability of distinguishing ρ from σ with a single measurement. We will use it to quantify the logical distinguishability: for a binary logic gate, the two possible output states $\rho_{\text{out}}^{(0)}$ and $\rho_{\text{out}}^{(1)}$ are perfectly distinguishable when $D = 1$, and completely indistinguishable when $D = 0$.

16 Architecture of the Floquet-buffered logic gate

The physical device is sketched in Fig. 32. It consists of:

- **Logical qubits** $\mathcal{S}$, described by a time-independent Hamiltonian H_S . For concreteness, in the numerical programme we will use three qubits forming a NOR gate as in [1].
- **Floquet buffer** $\mathcal{B}_F$, a finite quantum system with a time-periodic Hamiltonian $H_{\text{buf}}(t) = H_{\text{buf}}(t + T)$, $T = 2\pi/\Omega$. The buffer could be, for example, a harmonic oscillator with a modulated frequency or a driven two-level system.
- **System-buffer coupling** $H_{S\text{buf}}$, assumed time-independent and, for the weak-coupling theorem, weak.
- **Thermal baths**, each at a different temperature T_α , coupled only to the buffer via H_{buf,B_α} .

The logical input is encoded in the temperatures $\{T_\alpha\}$ of the baths, exactly as in the autonomous thermodynamic neuron. The output, however, is not read from an auxiliary bath; instead, we measure an observable of the output qubit stroboscopically at times $t_n = nT$, after a periodic steady state has been reached. This is a deliberate departure from strict autonomy: the device requires an external clock to synchronise the drive and the readout.

The reasons for this choice are twofold. First, in the autonomous model the output bath temperature depends on the qubit-bath coupling, which is absent in our architecture (the qubits couple only to the buffer). Second, stroboscopic readout is natural in Floquet systems and is used in many quantum computing platforms.

Because we accept a clock, the thermodynamics of the device is that of a periodically driven open system. The relevant energetic quantities are the cycle-averaged work $\overline{W}$ provided by the drive and the cycle-averaged heat currents $\overline{J}_\alpha$ into the baths.

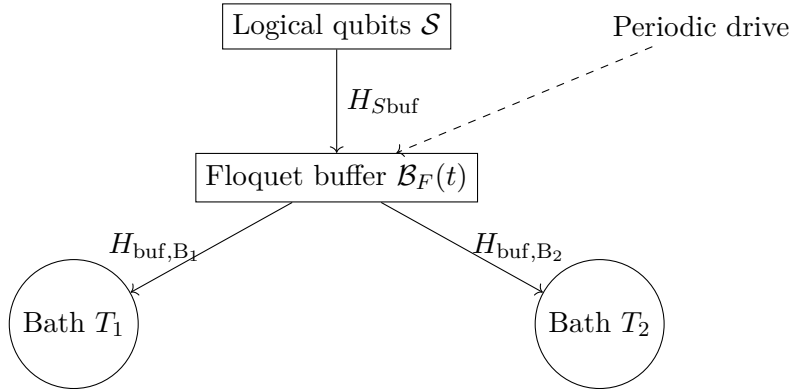


Figure 32: Periodically driven logic gate with Floquet buffer. The logical qubits interact only with the driven buffer; the buffer couples to thermal reservoirs. The device is not fully autonomous: a classical clock synchronises the drive and the stroboscopic readout.

17 Weak-coupling regime: periodic GKSL generator

Under certain idealised conditions, one can eliminate the buffer and the baths and obtain a closed description of the logical qubits alone.

17.1 Assumptions

We impose:

- (A1) **Weak buffer-bath coupling.** The interactions H_{buf, B_α} are sufficiently weak that the buffer, when decoupled from $\mathcal{S}$, relaxes to a unique periodic steady state $\rho_{\text{buf}, ss}(t)$ under the Floquet-Lindblad equation. The generator of this buffer-only dynamics is $\mathcal{L}_{\text{buf}}(t)$.
- (A2) **Exponential decay of buffer correlations.** In that periodic steady state, the two-time correlation functions $\langle B_i(t) B_j(t - \tau) \rangle$ decay to zero on a characteristic time scale τ_c . More precisely, there exists τ_c such that $|C_{ij}(t, \tau)| < \varepsilon$ for $\tau > \tau_c$ for any chosen tolerance ε .
- (A3) **Weak system-buffer coupling and time-scale separation.** The coupling $H_{S\text{buf}}$ is much weaker than the buffer's internal relaxation scale, and the logical qubits evolve slowly compared to τ_c . This allows us to apply the Born–Markov approximation.
- (A4) **Secular approximation.** The Bohr frequencies (transition frequencies) of H_S are non-degenerate and well separated compared to the relaxation rates induced by the buffer.

17.2 Periodic GKSL generator

Proposition 17.1. *Under Assumptions (A1)–(A4), there exists a time-periodic generator $\mathcal{L}_{\text{eff}}(t)$ of the form*

$$\mathcal{L}_{\text{eff}}(t)\rho = -i[H_S + H_{LS}(t), \rho] + \sum_k \Gamma_k(t) \left(L_k(t) \rho L_k^\dagger(t) - \frac{1}{2} \{ L_k^\dagger(t) L_k(t), \rho \} \right), \quad (90)$$

with $\mathcal{L}_{\text{eff}}(t + T) = \mathcal{L}_{\text{eff}}(t)$, such that the reduced state of the logical qubits obeys

$$\dot{\rho}_S(t) = \mathcal{L}_{\text{eff}}(t)\rho_S(t). \quad (91)$$

The jump operators $L_k(t)$ and rates $\Gamma_k(t)$ are obtained from the Fourier transforms of the buffer correlation functions evaluated at the Bohr frequencies of H_S , including sideband shifts by multiples of Ω .

Proof sketch. The derivation follows the standard Floquet–Born–Markov approach [9]. One starts from the Nakajima–Zwanzig equation with the buffer acting as a structured environment. The memory kernel is of second order in $H_{S\text{buf}}$. Because of (A2) the kernel decays on the time scale τ_c , and because of (A3) the system density matrix changes slowly on that scale; we can therefore make the Markov approximation and replace the memory integral by a time-local superoperator. The time dependence of this superoperator is inherited from the periodic buffer steady state. The secular approximation (A4) diagonalises the dynamics in the eigenbasis of H_S , yielding the GKSL form with time-periodic coefficients.

Practical limitations. The assumptions are strong. In particular, the weak-coupling condition (A3) may be hard to satisfy simultaneously with (A1) because weak buffer-bath coupling implies a long buffer correlation time, whereas a Markovian elimination requires the buffer correlations to decay fast *relative to the qubit dynamics*. Therefore Proposition 17.1 should be seen as identifying a parameter window in which a simple theoretical description exists, not as a statement about generic parameters.

A second issue is the dynamic Lamb shift. The Floquet buffer will induce time-dependent shifts $H_{LS}(t)$ in the qubit energy levels. The autonomous thermodynamic NOT gate relies on a precise resonance condition $\omega_W = \omega_X + \omega_Y$ among three qubit frequencies. The Floquet-induced Lamb shift will generically detune this resonance and destroy the logical operation. To mitigate this, we must actively adjust the bare qubit frequencies as functions of the drive amplitude, a procedure that is realistic in circuit-QED architectures but constitutes an additional control overhead.

For all these reasons, the main numerical programme of this project targets the regime *outside* the validity of Proposition 17.1, where no simple Lindbladian exists. Nevertheless, the weak-coupling analysis is a useful conceptual starting point.

18 Non-Markovian regime: chain mapping and MPS

To treat arbitrary coupling strengths and memory effects, we simulate the full system-buffer-baths compound as a closed many-body quantum system. We employ two established techniques: chain mapping of the baths (Sec. 15.3) and matrix-product state evolution (Sec. 15.4).

18.1 Chain-mapped Hamiltonian

Starting from the original bath model with a continuum of harmonic oscillators and a given spectral density $J(\omega)$, we apply a unitary transformation (Lanczos or orthogonal-polynomial mapping) that yields the chain Hamiltonian (87). The total Hamiltonian in the chain picture is

$$\begin{aligned} H_{\text{tot}}(t) = & H_S + H_{\text{buf}}(t) + H_{S\text{buf}} \\ & + \sum_{\alpha} \left[\sum_{k=1}^{N_{\alpha}} \omega_{\alpha k} a_{\alpha k}^{\dagger} a_{\alpha k} + \sum_{k=1}^{N_{\alpha}-1} t_{\alpha k} (a_{\alpha k}^{\dagger} a_{\alpha, k+1} + \text{h.c.}) \right] \\ & + \sum_{\alpha} V_{\text{buf}, \alpha} \otimes (a_{\alpha 1} + a_{\alpha 1}^{\dagger}). \end{aligned} \quad (92)$$

Here N_{α} is the chain length for bath α (chosen large enough to suppress finite-size effects), and $V_{\text{buf}, \alpha}$ is a buffer operator determined by the original buffer-bath coupling.

This Hamiltonian describes a one-dimensional lattice system. The logical qubits sit at the leftmost end, the buffer is next, and each bath forms a separate chain. The only interaction between qubits and baths is mediated by the buffer: the qubits couple to the buffer, and the buffer couples to the first site of each chain.

18.2 Energy current and work

Because the chain Hamiltonian is a closed system, thermodynamic observables can be defined exactly from the Heisenberg equations of motion.

The energy current from the logical qubits into the buffer is the time derivative of the qubit energy minus the contribution that would occur without the coupling:

$$J_{S \rightarrow \text{buf}}(t) = i \langle [H_S, H_{S\text{buf}}] \rangle. \quad (93)$$

Here $\langle \cdot \rangle = \text{Tr}(\rho_{\text{tot}}(t) \cdot)$, with ρ_{tot} the state of the full chain. Because H_S acts only on the qubits and $H_{S\text{buf}}$ acts on qubits and buffer, the commutator is a local operator with support on the qubits and the buffer alone. Its expectation value can therefore be computed directly from the MPS representation of the state without needing to reconstruct the full bath density matrix.

The power injected by the periodic drive is

$$P_{\text{drive}}(t) = \left\langle \frac{\partial H_{\text{buf}}(t)}{\partial t} \right\rangle. \quad (94)$$

This is also a local observable (it acts only on the buffer). The cycle-averaged work is

$$\overline{W} = \frac{1}{T} \int_0^T P_{\text{drive}}(t) dt. \quad (95)$$

The heat dumped into bath α can be obtained from the energy current flowing from the buffer into the first site of chain α , using the buffer-chain coupling. This current is again a local observable.

18.3 Bond-dimension control

A crucial practical question is whether the MPS simulation remains tractable. For a one-dimensional chain with nearest-neighbour couplings, the entanglement entropy of the state follows an area law: for a gapped system it saturates to a constant; for a critical (gapless) system it grows logarithmically with the chain length. The chain-mapped bath is gapless (the site frequencies ω_k extend to zero), but the coupling t_k decays rapidly with k , so that only the first few tens of sites are strongly entangled with the system.

In practice, for an Ohmic spectral density with a high-frequency cutoff and for temperatures $k_B T$ above the smallest energy scale of the chain, the bond dimension required for convergence is $\chi \approx 30$ –50. This value does *not* increase when the chain length N is increased beyond the point where the first few sites have captured the relevant physics. The computational cost per time step then scales as $O(N\chi^3 d^2)$, where d is the local Hilbert space dimension. For a truncated Fock space with $d = 10$ –20, this is manageable for the single-qubit buffer prototype (Phase 3 of the numerical programme) and, with tree-tensor-network (TTN) enhancements, for the three-qubit gate (Phase 4).

19 Conjectured work–distinguishability tradeoff

In the autonomous thermodynamic neuron, logical distinguishability is intrinsically linked to the temperature difference between the baths; increasing the distinguishability requires a larger temperature gradient, which implies larger entropy production. In our driven setting, we expect a similar tradeoff: improving the logical distinguishability beyond the undriven baseline should require a positive amount of work per cycle.

Formally, let

$$P_{\text{dist}} = \frac{1}{2} \left\| \rho_{\text{out}}^{(0)} - \rho_{\text{out}}^{(1)} \right\|_1 \quad (96)$$

be the trace-distance distinguishability of the output qubit states corresponding to the two logical inputs. Let $P_{\text{dist}}^{(0)}$ be this quantity when the buffer is undriven (i.e., the baseline autonomous gate, but with the same qubit readout), and let $\bar{W}$ be the cycle-averaged work when the buffer is driven.

Conjecture 19.1 (Work–distinguishability tradeoff). *There exists a monotonically increasing function f with $f(P_{\text{dist}}^{(0)}) = 0$ such that*

$$\bar{W} \geq f(P_{\text{dist}}). \quad (97)$$

In particular, any enhancement $P_{\text{dist}} > P_{\text{dist}}^{(0)}$ requires $\bar{W} > 0$.

This conjecture is motivated by known thermodynamic uncertainty relations (TURs) and speed-limit inequalities, which bound the precision of a process in terms of the entropy production or work. The precise functional form of f and its dependence on the bath temperatures are open questions that the numerical programme will explore.

20 Numerical programme

We propose a staged numerical investigation, designed to proceed from controlled to more challenging regimes.

Phase 1: Undriven baseline. Reproduce the autonomous NOT gate of [1] using QuTiP. However, instead of reading the output from a bath temperature, we read it from the reduced state of a designated output qubit. This gives us the undriven distinguishability $P_{\text{dist}}^{(0)}$ and the baseline steady-state heat currents.

Phase 2: Weak-coupling Floquet buffer (QuTiP). Implement a single qubit coupled to a driven harmonic oscillator (the buffer), which is weakly coupled to two Ohmic baths. Use QuTiP’s Floquet-Lindblad module to numerically verify Proposition 17.1. Explore how P_{dist} and $\bar{W}$ depend on drive amplitude and frequency. Check whether the conjecture holds in this regime.

Phase 3: MPS chain simulation (single qubit). Build the chain-mapped model of one qubit, one driven buffer, and two thermal chains using ITensor or MPSDynamics.jl. Initialise the chains in thermal states (via thermofield purification). Evolve in time until the periodic steady state is reached, then compute the energy current (93), the drive power (94), and the output qubit distinguishability. Investigate parameter regimes where the weak-coupling assumptions fail, and test the conjecture beyond weak coupling.

Phase 4: Three-qubit NOR gate. Extend the chain mapping to the full three-qubit NOR gate with a single Floquet buffer. Here the bond dimension may grow; if it becomes prohibitive, we will use tree tensor networks (TTNs) or systematic truncation of the least-coupled chain sites. The goal is to produce a phase diagram of P_{dist} as a function of drive amplitude and frequency, and to test the conjecture for a practical logic gate.

All code and data will be made publicly available to facilitate reproducibility.

21 Conclusion

We have presented a self-contained, pedagogical introduction to the Floquet-buffer architecture for periodically driven logic gates. The note reviews the necessary technical background, states a rigorous theorem in the weak-coupling limit, explains how to treat the non-Markovian regime with chain-mapped MPS, and formulates a clear conjecture to be tested numerically. The project is designed to utilise the complementary strengths of the collaboration: the physicist’s formal analysis, the mathematician’s tensor-network expertise, and the student’s numerical skills. This note is the definitive starting point for the work.

A Derivation of the GKSL equation from a microscopic model

For completeness we sketch the derivation of the GKSL equation (83) from a system-bath Hamiltonian. Consider a system S coupled to a bath B via $H_{SB} = \sum_i S_i \otimes B_i$. In the Born–Markov approximation, after tracing out the bath, one obtains the Redfield equation. The additional secular (rotating-wave) approximation diagonalises the dynamics in the eigenbasis of H_S and ensures complete positivity. The resulting jump operators are the eigenoperators $A(\omega)$ of the system Hamiltonian, with rates $\gamma(\omega) = \int e^{i\omega t} \langle B^\dagger(t) B(0) \rangle dt$. The detailed balance condition $\gamma(-\omega) = e^{-\beta\omega} \gamma(\omega)$ follows from the KMS condition of the bath. See [3] for a textbook treatment.

B Chain mapping: recurrence relations

Given a spectral density $J(\omega)$, the chain mapping is constructed by orthogonalising the polynomials in ω with respect to the measure $J(\omega)d\omega$. The recurrence coefficients ω_k and t_k are obtained from the continued fraction expansion of the bath correlation function. For the common case of an Ohmic bath with exponential cutoff, $J(\omega) = \eta\omega e^{-\omega/\omega_c}$, the coefficients can be computed analytically or numerically using the Lanczos algorithm [11].

C MPS representation of a thermal state

To initialise the chain in a thermal state at temperature T , one uses the thermofield (purification) approach. Each physical bosonic site is doubled into an ancilla site, and the initial state is the vacuum for a set of Bogoliubov-transformed modes. The resulting MPS bond dimension is directly related to the effective temperature; for moderate T , it remains small.

D Energy current operator: explicit form

Equation (93) follows from the Heisenberg equation for H_S :

$$\frac{dH_S}{dt} = i[H_{\text{tot}}, H_S] = i[H_S, H_{S\text{buf}}] + i[H_S, H_S] = i[H_S, H_{S\text{buf}}]. \quad (98)$$

The term $[H_S, H_S] = 0$, and H_S commutes with all bath and buffer-bath Hamiltonians because it acts on a different tensor factor. Hence the energy change of the qubits is solely due to the qubit-buffer interaction. Taking the expectation value gives the current.

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